

Financial Econometrics

1 Simple Linear Regression

1.1 Assumptions for OLS Estimation

Unbiasedness : $\mathbb{E}[\hat{\beta}] = \beta$

- Assumption 1 : Linear in the parameter, β
- Assumption 2 : Random Sampling
- Assumption 3 : No Multicollinearity
- Assumption 4 : $\mathbb{E}[u_i | X] = 0$

Efficiency :

$$\mathbb{E}[\omega_1] = \mathbb{E}[\omega_2] = \omega \quad \text{and} \quad \text{Var}(\omega_1) < \text{Var}(\omega_2)$$

Gauss-Markov Theorem : OLS is efficient in the class of linear unbiased estimators. To prove the Gauss-Markov theorem, we need Assumption 5.

- Assumption 5 : Homoskedasticity, $\text{Var}(u_i) = \sigma^2$

1.2 OLS Estimators

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (y_i - \bar{y})x_i}{\sum_{i=1}^n (x_i - \bar{x})x_i} = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sum_{i=1}^n y_i(x_i - \bar{x})}{\sum_{i=1}^n x_i(x_i - \bar{x})}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \quad \text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad \text{In MLR, } \text{Var}(\beta_1 | X) = \sigma^2 (X^T X)^{-1}$$

$$\sigma^2 = \frac{1}{n-2} \sum_{i=1}^n u_i^2 = \frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$R^2 = \frac{ESS}{TSS} = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2}, \quad TSS = ESS + RSS$$

2 Arbitrage Pricing

$$R_{it}^e = \alpha_i + \beta_{im} R_{mt}^e + \varepsilon_{it}$$

- $R_{it}^e = R_{it} - R_{ft} \equiv$ Excess return of asset i at time t
- $R_{mt}^e = R_{mt} - R_{ft} \equiv$ Excess market return
- ε_{it} = Idiosyncratic risk

$$\mathbb{E}[\varepsilon_{it} \varepsilon_{jt}] = 0 \quad \forall i \neq j$$

Residual risks are uncorrelated across assets. In other words, residual risk in one asset is idiosyncratic unrelated to the residual risk of another asset.

CAPM holds if $\alpha_i \approx 0$. Now consider a portfolio :

$$R_{pt}^e = \alpha_p + \beta_{pm} R_{mt}^e + \varepsilon_{pt}$$

$$\alpha_p = \sum_j w_j \alpha_j, \quad \beta_{pm} = \sum_j w_j \beta_{jm}, \quad \varepsilon_{pt} = \sum_j w_j \varepsilon_{jt}$$

If $w_j = \frac{1}{N}$, then $\text{Var}(\varepsilon_{pt}) = \frac{\bar{\sigma}^2}{N} \rightarrow 0$ as $N \rightarrow \infty$

Also, $\text{Var}(\varepsilon_{pt}) = \sum_j w_j^2 \text{Var}(\varepsilon_{jt})$. For large N , $\alpha_p \approx 0$.

If $\alpha_p \neq 0$, there is an **arbitrage opportunity** : long the portfolio and short β_{pm} units of the market.

Ross Intuition :

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum \alpha_i^2 = 0$$

A few assets may be mispriced, but if many are mispriced they can be combined into a diversified portfolio generating arbitrage.

Suppose we invest $\frac{\alpha_i}{\sqrt{N \sum \alpha_i^2}}$ dollars in asset i . The portfolio payoff is $\frac{\sum \alpha_i R_{it}}{\sqrt{N \sum \alpha_i^2}}$.

Using $R_{it} = R_{ft} + \alpha_i + \beta_{im} R_{mt}^e + \varepsilon_{it}$, we get,

$$\frac{\sum \alpha_i R_{it}}{\sqrt{N \sum \alpha_i^2}} = \frac{\sum \alpha_i R_{ft}}{\sqrt{N \sum \alpha_i^2}} + \frac{\sqrt{\sum \alpha_i^2}}{\sqrt{N}} + \frac{\sum \alpha_i \beta_{im}}{\sqrt{N \sum \alpha_i^2}} R_{mt}^e + \frac{\sum \alpha_i \varepsilon_{it}}{\sqrt{N \sum \alpha_i^2}}$$

- First term $\rightarrow 0$
- Second term = target return
- Third term = systematic part
- Fourth term = noise

If there is a systematic relation between alphas and betas in a large cross-section of assets with uncorrelated residual risk, say a negative relation, one can make riskless profits by :

- Going long low-beta assets
- Going short high-beta assets

3 Asset Pricing

CAPM Framework : $R_{it}^e = \alpha_i + \beta_{im} R_{mt}^e + \varepsilon_{it}$

Null Hypothesis : $H_0 : \alpha_1 = \alpha_2 = \dots = \alpha_N = 0$

Time-Series Approach

For each asset i , estimate the time-series regression and obtain $\hat{\alpha}_i$.

Large T :

$$T \left[1 + \left(\frac{\bar{R}_m^e}{\hat{\sigma}(R_{mt}^e)} \right)^2 \right]^{-1} \hat{\alpha}' \hat{\Omega}^{-1} \hat{\alpha} \sim \chi_N^2$$

$\hat{\Omega} \equiv$ Covariance matrix of ε_{it}

Finite T :

$$\left(\frac{T - N - 1}{N} \right) \left[1 + \left(\frac{\bar{R}_m^e}{\hat{\sigma}(R_{mt}^e)} \right)^2 \right]^{-1} \hat{\alpha}' \hat{\Omega}^{-1} \hat{\alpha} \sim F_{N, T-N-1}$$

Reject H_0 if the test statistic exceeds the critical value. Reject $H_0 \Rightarrow$ CAPM doesn't hold in the data.

Cross-Sectional Approach

Step 1 : Estimate $\hat{\beta}_{im}$ from time-series regressions.

Step 2 : Run the cross-sectional regression with the averages of the returns to get λ :

$$\bar{R}_i^e = \lambda \hat{\beta}_{im} + a_i \quad \bar{R}_i^e = \frac{1}{T} \sum_{t=1}^T R_{it}^e$$

The interpretation of λ is the price of risk. If CAPM is true then a_i is 0.

Consistency (as $T \rightarrow \infty$) :

$$\bar{R}_i^e \xrightarrow{p} E[R_{it}^e], \quad \hat{\beta}_{im} \xrightarrow{p} \beta_{im}$$

$$a_i \xrightarrow{p} \alpha_i, \quad \lambda \xrightarrow{p} E[R_{mt}^e]$$

The cross-sectional residual a_i is a constant estimator of α_i . Hence,

$$E[R_{it}^e] = \alpha_i + \beta_{im} E[R_{mt}^e]$$

GLS Estimation :

$$\hat{\lambda}_{GLS} = (\hat{\beta}_m' \hat{\Omega}^{-1} \hat{\beta}_m)^{-1} \hat{\beta}_m' \hat{\Omega}^{-1} \bar{R}^e$$

$$\hat{\alpha}_{GLS} = \bar{R}^e - \hat{\lambda}_{GLS} \hat{\beta}_m$$

Test Statistic :

$$T \left[1 + \left(\frac{\hat{\lambda}_{GLS}}{\hat{\sigma}(R_{mt}^e)} \right)^2 \right]^{-1} \hat{\alpha}_{GLS}' \hat{\Omega}^{-1} \hat{\alpha}_{GLS} \sim \chi_{N-1}^2$$

One degree of freedom is lost due to estimation of λ .

Fama-MacBeth Procedure

Step 1 : Estimate $\hat{\beta}_{im}$.

Step 2 : For each t , run :

$$R_{it}^e = \lambda_t \hat{\beta}_{im} + a_{it}$$

Obtain $\hat{\lambda}_t$ and $\hat{a}_{it}$.

Average Estimates :

$$\hat{\lambda}_{FM} = \frac{1}{T} \sum_{t=1}^T \hat{\lambda}_t \quad \hat{a}_{i,FM} = \frac{1}{T} \sum_{t=1}^T \hat{a}_{it}$$

Variance Estimation :

$$\widehat{\text{Var}}(\hat{\lambda}_{FM}) = \frac{1}{T^2} \sum_{t=1}^T (\hat{\lambda}_t - \hat{\lambda}_{FM})^2$$

$$\widehat{\text{Var}}(\hat{a}_{i,FM}) = \frac{1}{T^2} \sum_{t=1}^T (\hat{a}_{it} - \hat{a}_{i,FM})^2$$

Testing : In this approach, we can use a test statistic instead of the complex distributions mentioned earlier :

$$t = \frac{\hat{\alpha}_i}{\sqrt{\text{Var}(\hat{\alpha}_i)}} \sim N(0, 1)$$

4 Firm Characteristics and Cross section of Expected Utility Returns by Fama French

Fama-MacBeth Procedure : Average Effect of Characteristic

The procedure utilizes the following regression model :

$$r_{it} = a_t + \delta_i C_{i,t-1} + \varepsilon_{it}$$

- r_{it} : represents **excess returns**.
- $C_{i,t-1}$: represents information such as the **Book to market ratio** or any **leverage ratio**. This characteristic can be **time constant**.

Methodology and Hypothesis Testing

The approach uses time series δ_t to compute the average effect and perform hypothesis testing :

- Average Effect :** $\bar{\delta} = \frac{1}{T} \sum_{t=1}^T \hat{\delta}_t$

- Variance :** $\text{Var}(\bar{\delta}) = \frac{1}{T^2} \sum (\hat{\delta}_t - \bar{\delta})^2 = \frac{1}{T} \cdot \text{Var}(\hat{\delta})$

- Hypothesis :** $H_0 : \bar{\delta} = 0$

- Test Statistic :**

$$t = \frac{\bar{\delta}}{\sqrt{\text{Var}(\bar{\delta})}} \sim N(0, 1)$$

Analysis Conclusions (using $\ln(\text{Market Equity})$)

- $\bar{\delta}$ is the **average effect** of a characteristic.
- The effect does not need to be present at every time t , but should be present **most of the time** if $\bar{\delta}$ is significant.
- A characteristic cannot be in an asset pricing model.

Findings on CAPM and Stock Returns

- The **CAPM model** and **market β** are deemed **not adequate** or useful in real-world applications.
- The cross section of stock returns is instead explained by valuation ratios like **Book to market, leverage, and price earnings ratio**.
- **Size** is an important driver of the cross section. Fama and French noted that once size is corrected for, **β fails to explain anything** on the cross-section.
- The **Book to market ratio** is a significant driver that can **absorb the impact** of other characteristics :

$$\frac{BE}{ME} > 1 : \text{Value firm (Buy)}$$

$$\frac{BE}{ME} < 1 : \text{Growth firm (Sell)}$$

- Estimating excess returns of the cross section does not require the use of many characteristics.

5 Momentum

Momentum is a trading strategy based on **buying past winners and selling past losers**.

- The momentum effect is **large and robust** ; it cannot be explained by factors like market size or value.
- Momentum is considered an “**asset**” **characteristic**, whereas size, value, investment, and profitability are “firm” characteristics.
- It is **not a risk-based factor**. Because profits can be generated as long as there is relative pricing (even during “hard times”), it is not a risk-linked parameter.

Jagadeesh & Titman One-Factor Model

The model for returns on asset i, is defined as :

$$r_{it} = \mu_i + \beta_i f_t + e_{it}$$

- f_t : Some common shock factor that is not fixed, it can be a linear combination of several factors to design a one-factor model. It is common for all stocks, for ex, f_t can be excess market return ($r_m - r^f$) or systematic factor return.
- e_{it} : Idiosyncratic error, where
$$\mathbb{E}[e_{it}] = 0, \quad \mathbb{E}[e_{it}e_{jt}] = 0, \quad \mathbb{E}[e_{it}f_t] = 0.$$
- $\mu_i = r^f + \beta_i \lambda$: The expected return on asset i , where r^f is the risk-free rate and λ is the risk premium.
- $\mathbb{E}[f_t] = 0$, The factor is normalized for analysis.
- **Portfolio Classification** : High μ_i indicates a winner portfolio, while lower μ_i indicates a loser portfolio.

f_t is the unexpected shock factor and in the Fama-French model it is market excess returns (MKT), HML (High Minus Low), SMB (Small Minus Big), and Momentum. Each one represents a systematic return factor observed in the data. λ is the risk price. It is the expected compensation for exposure to that shock. In CAPM,

$$\lambda = \mathbb{E}[r_m - r_f]$$

The first model :

$$r_{it} = \mu_i + \beta_i f_t + e_{it}$$

- This is a **time-series model**.
- It explains **actual realized returns in a given period t** as a combination of :
 - μ_i = expected return of stock i
 - $\beta_i f_t$ = exposure to a **factor shock** that varies over time
 - e_{it} = idiosyncratic noise

Today, stock i returned this much. Part is expected, part is due to systematic factor movements, and part is random noise.

- This is **observable** because r_{it} and f_t are known at time t .

The second model :

$$\mu_i = r^f + \beta_i \lambda$$

- This is a **cross-sectional or equilibrium model**.
- It describes **expected returns across assets** :

Expected return of stock i = risk-free rate + factor exposure $\times$ price of risk

- λ = price of factor risk (how much extra return investors demand for taking factor risk)
- μ_i = the **time-series mean of r_{it}**
- This is **not directly observable in one period** ; it is a theoretical construct about **average expected returns**.

Momentum in Theory (Autocovariance)

The theoretical basis for momentum is positive autocovariance :

$$\mathbb{E}[(r_{it} - \bar{r}_t) \mid r_{it-1} - \bar{r}_{t-1} > 0] > 0$$

$$\mathbb{E}[(r_{it} - \bar{r}_t) \mid r_{it-1} - \bar{r}_{t-1} < 0] < 0$$

This implies :

$$\mathbb{E}[(r_{it} - \bar{r}_t)(r_{it-1} - \bar{r}_{t-1})] = (\mu_i - \bar{\mu})^2 + (\beta_i - \bar{\beta})^2 \text{cov}[f_t, f_{t-1}] + \text{cov}[e_{it}, e_{it-1}]$$

By averaging across all stocks, we get :

$$\sigma_\mu^2 + \sigma_\beta^2 \text{cov}[f_t, f_{t-1}] + \text{cov}[e_{it}, e_{it-1}]$$

Drivers of the Momentum Effect

1. **Large unconditional difference in expected returns (σ_μ^2)** leads to a large momentum effect.
2. **Large autocovariance or large dispersion in asset sensitivities (σ_β^2)** leads to a large momentum effect.
3. $\sigma_\beta^2 \text{cov}[f_t, f_{t-1}]$ interpretation :
 - If $f_{t-1} > 0$: Stocks with high β are winners and expected to be winners at t ; stocks with low β are losers and expected to be losers at t .
 - If $f_{t-1} < 0$: Stocks with high β are losers and expected to be losers at t ; stocks with low β are winners and expected to be winners at t .

Observations on Factors and Residuals

- Analysis of factors like **MKT, SMB, and HML** shows that their autocovariance **does not explain momentum**.
- The **autocovariance of residuals** may explain momentum, specifically positive autocovariance in pricing errors.

6 Common Risk Factors in Returns by Fama and French

The Capital Asset Pricing Model (CAPM) predicts that the market portfolio alone prices all assets and does not take into account of traded assets like bonds, commodities and & non-traded assets like warehouses, human capital etc. The CAPM cannot account for the well-documented size effect and value effect, motivating additional systematic risk factors. So, we need additional factors, more systematic factors to capture the variance in the returns. CAPM is also considered as non scientific method and rejected with equity market portfolio. Those additional factors are firm characteristics.

6.1 Creating a factor

A systematic factor is the return on a long-short portfolio. It is a procedure to move from a firm level characteristic to a tradable factor.

- **Step 1** : Use the Fama-MacBeth to identify significant predictors. Run period-by-period cross-sectional regressions of individual excess returns on lagged characteristic $c_{i,t-1}$. Average the time-series of slope estimates ; statistical significance flags a useful trading signal.
- **Step 2** : Sort all stocks into m **portfolios** based on $c_{i,t-1}$. Common choices : quintile ($m = 5$) or decile ($m = 10$), depending on the number of available stocks.

- **Step 3** : Hold the sorted portfolios for a fixed horizon (typically one year), record portfolio returns, then **re-sort and reform**. Repeat over the entire sample.
- **Step 4** : The **factor return** is the return spread between the top (long) and bottom (short) portfolio, a self-financing long-short position.

We can follow equally weighted or value weighted schemes for the portfolios. If we implicitly overweights small, illiquid firms whose paper returns are hard to realise. Value weighted by market capitalisation, giving more influence to large, liquid firms. Preferred in practice because transaction costs are lower and returns are achievable.

- The factor is a **traded asset**, so its risk premium equals its time-series average return, which is the best unbiased estimator, so no cross-sectional regression needed.
- **Factor sensitivities (betas)** are estimated by time series regression of asset excess returns on the factor return, which is free of errors in varaibles bias.
- The factor is common to all stocks ($F_{k,t}$) distinguishing a true systematic factor from a firm specific characteristic.

The multi factor model,

$$E[r_{it}] = \text{alpha}_i + \text{beta}_{1i} \times E[F_{1t}] + \text{beta}_{2i} \times E[F_{2t}] + \dots + \text{beta}_{Ki} \times E[F_{Kt}]$$

The Fama-French Three factor model,

$$r_{i,t} - R_{f,t} = \alpha_i + \beta_i (R_{m,t} - R_{f,t}) + s_i \text{SMB}_t + h_i \text{HML}_t + \varepsilon_{i,t}$$

- **Value stock** : High BE/ME, prices rises, expected higher returns
- **Growth stock** : Low BE/ME, slower future gains, lower expected returns.
- **SMB** : Average return of (S/L + S/M + S/H) minus average return of (B/L + B/M + B/H). Captures the size premium small stocks tend to earn higher average returns than large stocks.
- **HML** : Average return of (S/H + B/H) minus average return of (S/L + B/L). Captures the value premium high-BE/ME stocks earn higher average returns than growth stocks.
- **RM-RF** : Value-weighted return on all stocks in the six portfolios (plus negative-book-equity stocks) minus the one-month T-bill rate RF.

6.2 Fama-French Results

- **Size Effect** : Within each BE/ME quintile, average returns decrease as size grows from Small to Big.
- **Value Effect** : Within each size quintile, average returns increase as BE/ME grows from Low to High.

6.2.1 CAPM (Market Factor Only)

- Market betas : near-uniform across all 25 portfolios, clustering around 1. Because there is almost no spread in beta, the CAPM cannot generate the large spread in expected returns its core failure.
- R-squared : relatively high for large-cap / low-BE/ME portfolios (0.890.92) but much lower for small-cap portfolios (0.610.70), reflecting the value-weighting of the market factor.
- Alphas (intercepts) : significant patterns remain decrease from small to large, increase from low to high BE/ME. The CAPM systematically misprices small and value stocks.

6.2.2 SMB and HML Only

- SMB loading (s) : strongly decreasing from small (large positive) to large (near-zero or negative). Reflects the long-small position in SMB.
- HML loading (h) : strongly increasing from low-BE/ME (negative) to high-BE/ME (large positive). Reflects the long-value position in HML.
- R-squared : improves notably for small-cap portfolios where the market factor was weakest SMB and HML explain over 60% of variation for small firms.

6.2.3 All Three Factors Combined Key Results

- R-squared exceeds 90% for almost all 25 portfolios. The model explains the overwhelming majority of time-series variation.
- Alphas reduced to near-zero for nearly all portfolios, with $|t| < 2$. Virtually no abnormal return remains unexplained.
- Factor loadings are intuitive : s is large positive for small firms, negative for large and h is large positive for value stocks, negative for growth stocks.

6.2.4 Out-of-Sample Test : Earnings-to-Price Portfolios

Applying the three-factor model to 5 portfolios sorted on E/P ratio a characteristic not used in constructing the factors :

- Market factor only : significant alpha patterns (0.20 to +0.46), with $|t| > 2$ for several portfolios. The CAPM misprices E/P-sorted portfolios.
- Three-factor model : all alphas small and insignificant. The model successfully prices even these out-of-sample portfolios, demonstrating genuine pricing power.

In conclusion, three factors collectively explain a very high proportion of cross-sectional and time-series variation in stock returns. For bonds, two separate factors (TERM and DEF) explain bond returns. Stock and bond markets are largely segmented and stock factors matters only for low-grade bonds. The three factor become a standard dramtically reducing the pricing errors of CAPM.

- **Value of stock** : $M_t = \sum_{j=1}^{\infty} E(d_{t+j})/(1+r)^j$, sum of discounted expected dividends.
- **Dividends** : $d_t = Y_t - (B_t - B_{t-1}) = Y_t - dB_t$ Y_t : equity earnings $\approx$ profit, $dB_t = B_t - B_{t-1}$: change in book value $\approx$ investment
- Then value of the firm $M_t = \sum_{j=1}^{\infty} E[Y_{t+j} - dB_{t+j}]/(1+r)^j$ sum of discounted expected profits minus investments.

$$\frac{M_t}{B_t} = \frac{\sum_{j=1}^{\infty} E[Y_{t+j} - dB_{t+j}]/(1+r)^j}{B_t}$$

- Relation between discount rate r, market-to-book ratio, expected earning and investments.

7 Jump Diffusion Process

We usually focus on models for daily asset log-returns, $r_t = \Delta \log(P_t)$. r_t is return on an investment buying asset at end of yester and selling at the end of today. P_{t_0} is opening price on day t, $r_{t_i} = \log(P_{t_i}/P_{t_{i-1}})$, $i = 1, ..., n$ may be used to estimate variance of daily return $r_t = \sum_{i=1}^n r_{t,i}$

Let X be the stochastic process for the log-price of an asset. The dynamics of X are given by the Jump-Diffusion process discussed previously

$$X_t = \int_0^t \mu_s ds + \int_0^t \sqrt{c_s} dW_s + \tilde{Y}_t$$
$$dX_t = \mu_t dt + \sqrt{c_t} dW_t + d\tilde{Y}_t$$

The first term is drift part, second term is diffusion part and the third part is the Jump component. Simulation of X is breakdown into three parts, simulating the diffusion, simulating the jump and combining both.

7.1 Constant Variance

- Say, $\sqrt{c} = \sigma$, for some $\sigma \in R$, if $\sigma = 0.011$, then it is 1.10% per day.
- Discretization interval is $\Delta_n = \frac{1}{n}$. Say no.of steps per day is 78, then $\Delta = 1/78$
- No.of trading day in a year = 252
- Assume Drift is constant and equal to zero

7.1.1 Continuous Part

$\tilde{X}$ is the diffusion part of the process. Generate nT independent draws from standard normal distribution $Z_i \sim \mathcal{N}(0, 1) \quad i = 1, 2, 3, ..., nT$.

$$\tilde{X}_i = \tilde{X}_{i-1} + \mu \Delta_n + \sigma \sqrt{\Delta_n} Z_i, \quad \text{for } i = 1, 2, ..., nT$$

Take $X_0 = \tilde{X}_0 = \log(initial\ price)$. After computing all the $\tilde{X}_i$'s, convert back to prices by $\tilde{P}_i = e^{\tilde{X}_i}$

7.1.2 The Jump Part

- Define the intensity of jumps, say $\lambda = 100/252$ (100 jumps per year on average), intensity is the frequency at which jumps occur, i.e jumps per unit of time.Intensity ($\lambda = 2/252$) : "On average, I expect about 2 shocks this year", Number of Jumps (N=100) : "I actually experienced 100 shocks this year".

- Obtain total no.of jumps by taking a random draw from a poisson distribution $N \sim Poisson(\lambda T)$. Draw N jumps from a normal distribution with variance, σ_{jump}^2

$$Y_k \sim \mathcal{N}(0, \sigma_{jump}^2) \quad \text{for } k = 1, ..., N$$

- To distribute the jumps at random times, draw N independent random numbers from uniform distribution,

$$U_k \sim Unif(0, 1) \quad k = 1, 2, 3, ..., N$$

- Form the discretized jump process

$$J_i = \sum_{k=1}^N \mathbf{1}_{\{U_k \leq \frac{i}{nT}\}} Y_k \quad \text{for } i = 1, 2, ..., nT$$

$$\mathbf{1}_{\{U_k \leq \frac{i}{nT}\}} = \mathbf{1}_{\{TU_k \leq \frac{i}{n}\}} = \mathbf{1}_{\{TU_k \leq i \Delta_n\}}$$

The indicator is checking whether the realization of TU_k , uniform distribution over [0,T] is smaller than the position of i-th interval.

- J_i the accumulated total of all jump sizes that have occurred up until the i-th time step. It can be negative as well, if the size of the shocks are downward, then the accumaltion of shocks will be negative.
- N is the total number of jumps drawn from a Poisson distribution.
- Y_k represents the random size of the k -th jump.

7.1.3 Combining the Two parts

Continuous part $\tilde{X}_i$ s, simulation of jumps process J_i s. Combine both to get log-price process,

$$X_i = \tilde{X}_i + J_i \quad \text{Prices : } P_i = e^{X_i}$$

7.2 Stochastic variance

Now variance is time varying and random. Drift is dropped for convenience.

$$dX_t = \sqrt{c_t} dW_t \quad dc_t = \rho(\mu_c - c_t)dt + \sigma_c \sqrt{c_t} dW_t^c$$

W, W^c are independent. We will simulate the model at very high frequency and sample at a coarser frequency to mimic the actual observed data, to minimize the discretization error. For ex, model simulation frequency will be 15 sec, then sampling the simulation will be say 5 mins.

$$\text{Discretization Interval Size, } \delta_E = \frac{1}{n_E} < \Delta_n$$

- First generate n_ET independent draws from the standard normal distribution $\tilde{Z}_j \sim N(0, 1) \quad \text{for } j = 1, 2, ..., n_ET$
- Then, given an initial value for the variance ($c_0 = \mu_c$) we can iterate the variance process

$$c_j = c_{j-1} + \rho(\mu_c - c_{j-1})\delta_E + \sigma_c \sqrt{c_{j-1}} \sqrt{\delta_E} \tilde{Z}_j$$

There is a small chance that the simulated c_j could go negative. When that happens, set $c_j = \mu_c/2$

- Then, generate n_ET independent draws from the standard normal distribution

$$Z_j \sim N(0, 1) \quad \text{for } j = 1, 2, ..., n_ET$$

Note that $\tilde{Z}_j$ and Z_j are independent

- Now, given $\tilde{X}_0 = X_0 = \log 75$, we can iterate the process

$$\tilde{X}_j = \tilde{X}_{j-1} + \sqrt{\tilde{c}_{j-1}} \sqrt{\delta_E} Z_j \quad \text{for } j = 1, 2, ..., n_ET$$

- Finally, sample the $\tilde{X}_j$ at a coarser frequency

$$X_i = \tilde{X}_{(\frac{n_E}{n})} \quad \text{for } i = 1, 2, ..., nT$$

8 Estimating the Integrated variance

One of the primary uses of high-frequency financial data is to estimate the **integrated variance** (IV) of an asset over a trading day. The integrated variance is defined as :

$$IV = \int_0^1 c_s ds$$

where c_s is the *local variance process* (also written σ_s^2) and the interval [0, 1] represents a single trading day.

Why do we want to estimate this ? Knowing the daily variance is essential for tasks like computing the probability that a stock drops by more than 3% tomorrow, constructing derivative prices, or calibrating risk models. Traditionally, parametric models such as ARCH or GARCH were used. However, high-frequency data provides a fundamentally new and powerful approach :

- **More information** : Sampling at higher frequencies provides many more observations within a day, leading to more accurate variance estimates.
- **Non-parametric** : The estimators described here do not require specifying a model order (unlike ARCH/GARCH), so there is no risk of model misspecification bias.
- **Simplicity** : The estimators are computed by straightforward summations of squared returns-no maximum likelihood optimisation is required.

If you use 5-minute returns (giving $n = 78$ intervals per 6.5-hour trading day), the realized variance estimator is already an excellent proxy for the true daily variance. You gain accuracy simply by sampling more finely.

8.1 The Price Process : Setup

8.1.1 The Continuous Semimartingale Model

Let X_t denote the log-price process. In the continuous (no-jump) case :

$$X_t = X_0 + \int_0^t a_s ds + \int_0^t \sqrt{c_s} dW_s$$

- The first term $\int_0^t a_s ds$ is the **drift component**-an ordinary (Lebesgue) integral.
- The second term $\int_0^t \sqrt{c_s} dW_s$ is the **diffusive component**-a stochastic integral driven by a standard Brownian motion W_s .
- c_t is the **local variance process**, which is itself stochastic and reflects economic conditions (e.g., news arrivals, market regimes).

At high sampling frequencies, the drift term a_t is **negligible**.

- **Drift contribution** : $\int_t^{t+\Delta_n} a_s ds \approx a_t \cdot \Delta_n$. The magnitude is *of order* Δ_n .
- **Diffusive contribution** : $\int_t^{t+\Delta_n} \sqrt{c_s} dW_s$. Since Brownian motion has quadratic variation equal to time, the standard deviation of this term is *of order* $\sqrt{\Delta_n}$.

At high frequencies, the log-price process effectively reduces to :

$$X_{t+\tau} \approx X_t + \sqrt{\int_t^{t+\tau} c_s ds} \cdot Z_t, \quad Z_t \sim N(0, 1)$$

The approximation above uses the fact that a stochastic integral with a Brownian motion is a zero-mean, square-integrable martingale, and by Itô's isometry :

$$\text{Var} \left(\int_t^{t+\tau} \sqrt{c_s} dW_s \right) = E \left[\int_t^{t+\tau} c_s ds \right]$$

Hence the stochastic integral is approximately $N \left(0, \int_t^{t+\tau} c_s ds \right)$ over short intervals, since c_s changes slowly.

8.1.2 Discrete Observations and Returns

We observe log-prices at equally spaced times $0, \Delta_n, 2\Delta_n, \dots, n\Delta_n$ where $\Delta_n = 1/n$ and $n = \lfloor 1/\Delta_n \rfloor$. The **log return** over the i -th interval is :

$$\Delta_i^n X = X_{i\Delta_n} - X_{(i-1)\Delta_n}, \quad i = 1, \dots, n$$

Using the approximation above (assuming $c_s \approx c_{n,i}$ is locally constant on $[(i-1)\Delta_n, i\Delta_n]$) :

$$\Delta_i^n X \approx \sqrt{\bar{c}_{n,i}} Z_{n,i}, \quad Z_{n,i} \stackrel{iid}{\sim} N(0, 1)$$

where $\bar{c}_{n,i} = \int_{(i-1)\Delta_n}^{i\Delta_n} c_s ds \approx c_{n,i} \cdot \Delta_n$, and $c_{n,i}$ is the value of c somewhere in the i -th interval.

9 The Realized Variance : Definition and Properties

The **realized variance** (RV) for a single trading day ($t = 1$) is :

$$RV = \sum_{i=1}^n |\Delta_i^n X|^2$$

It is simply the sum of squared intraday log-returns.

In the multi-day notation used later, for day t with intraday returns $r_{t,i}$:

$$RV_t = \sum_{i=1}^n r_{t,i}^2$$

9.1 Law of Large Numbers : Consistency of RV

9.1.1 Constant Variance Case

Suppose $c_s = c$ is constant. Then :

$$RV \approx \sum_{i=1}^n |\sqrt{\Delta_n} c Z_i|^2 = \Delta_n c \sum_{i=1}^n |Z_i|^2 = c \cdot \frac{1}{n} \sum_{i=1}^n |Z_i|^2$$

By the standard Law of Large Numbers (LLN), $\frac{1}{n} \sum_{i=1}^n Z_i^2 \rightarrow E[Z^2] = 1$. Therefore :

$$RV \rightarrow c = \int_0^1 c ds = IV$$

The key insight is that the factor $\frac{1}{n}$ needed for the LLN is already embedded in $\Delta_n = \frac{1}{n}$.

9.1.2 Time-Varying Variance : Triangular Array LLN

When c_t is not constant, we need the **LLN for triangular arrays**. This states : for random variables $Y_{n,i}$ that are independent across i (for each fixed n) :

$$\frac{1}{n} \sum_{i=1}^n Y_{n,i} \rightarrow \mu, \quad \text{where } \mu = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n E[Y_{n,i}]$$

The term “triangular array” refers to the two-dimensional index structure :

$$Y_{1,1}; \quad Y_{2,1}, Y_{2,2}; \quad Y_{3,1}, Y_{3,2}, Y_{3,3}; \quad \dots$$

For each row (fixed n), the $Y_{n,i}$ need only be independent of each other-they need not be identically distributed, and they can be correlated across rows.

Setting $Y_{n,i} = c_{n,i} Z_{n,i}^2$, we can write :

$$RV = \frac{1}{n} \sum_{i=1}^n c_{n,i} Z_{n,i}^2$$

Then :

$$\frac{1}{n} \sum_{i=1}^n E[Y_{n,i}] = \Delta_n \sum_{i=1}^n c_{n,i} \rightarrow \int_0^1 c_s ds = IV$$

by the definition of the Riemann integral.

LLN for Realized Variance (Continuous Case) : As $\Delta_n \rightarrow 0$ (i.e., $n \rightarrow \infty$) :

$$RV \xrightarrow{p} IV = \int_0^1 c_s ds$$

This holds whether c_t is constant or time-varying, as long as the process is continuous (no jumps).

9.1.3 What Happens with Jumps ?

If X_t has jumps, the LLN result changes. Let $(\tau_p)_{p \geq 1}$ index the jump times. Then :

$$RV \xrightarrow{p} IV + \sum_p |\Delta_{\tau_p} X|^2$$

That is, RV converges to the integrated variance *plus* the sum of all squared jump sizes. This **jump contamination** is why alternative estimators are needed when jumps are present.

10 The Central Limit Theorem for RV

Knowing that $RV \rightarrow IV$ is not sufficient for inference. We also need the **rate of convergence** and the **limiting distribution** to form confidence intervals and conduct hypothesis tests.

10.1 The Convergence Rate

The standard convergence rate in statistics is $\sqrt{n}$: the error in estimating a mean from n observations is $O(n^{-1/2})$. The RV estimator achieves this optimal rate. Specifically, the relevant quantity is :

$$\Delta_n^{-1/2} \left(\sum_{i=1}^n |\Delta_i^n X|^2 - \int_0^1 c_s ds \right)$$

Since $\Delta_n = 1/n$, this equals $\sqrt{n}(RV - IV)$.

10.2 Derivation of the CLT

Using the approximation $\Delta_i^n X \approx \sqrt{\bar{c}_{n,i}} Z_{n,i}$:

$$\sum_{i=1}^n |\Delta_i^n X|^2 - \int_0^1 c_s ds \approx \sum_{i=1}^n \bar{c}_{n,i} (Z_{n,i}^2 - 1)$$

(since $E[Z_{n,i}^2] = 1$, so $\bar{c}_{n,i} Z_{n,i}^2 - \bar{c}_{n,i} = \bar{c}_{n,i} (Z_{n,i}^2 - 1)$).

Multiplying by $\Delta_n^{-1/2}$:

$$\Delta_n^{-1/2} \sum_{i=1}^n \bar{c}_{n,i} (Z_{n,i}^2 - 1)$$

10.2.1 Constant Variance Case

If $c_s = c$, then $\bar{c}_{n,i} = c \Delta_n$, so :

$$\Delta_n^{-1/2} \sum_{i=1}^n c \Delta_n (Z_{n,i}^2 - 1) = \frac{c}{\sqrt{n}} \sum_{i=1}^n (Z_{n,i}^2 - 1)$$

Let $Y_i = Z_{n,i}^2 - 1$. Then $E[Y_i] = 0$ and $\text{Var}(Y_i) = E[Z^4] - (E[Z^2])^2 = 3 - 1 = 2$.

By the standard CLT, $\frac{1}{\sqrt{n}} \sum_{i=1}^n Y_i \rightarrow N(0, 2)$. Since we multiply by c , the result is :

$$\frac{c}{\sqrt{n}} \sum_{i=1}^n (Z_{n,i}^2 - 1) \rightarrow N(0, 2c^2)$$

10.2.2 Time-Varying Variance Case

For general c_t , we use the **CLT for triangular arrays**. The term

$\Delta_n^{-1/2} \bar{c}_{n,i} (Z_{n,i}^2 - 1)$ plays the role of $Y_{n,i}$. The asymptotic variance is :

$$2 \times \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\Delta_n^{-1/2} \bar{c}_{n,i} \right)^2$$

Using $\bar{c}_{n,i} \approx c_{n,i} \Delta_n$:

$$2 \times \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\Delta_n^{-1/2} c_{n,i} \Delta_n \right)^2 = 2 \times \lim_{n \rightarrow \infty} \sum_{i=1}^n c_{n,i}^2 \Delta_n \rightarrow 2 \int_0^1 c_s^2 ds$$

again by the Riemann integral definition.

10.3 The CLT Result

Central Limit Theorem for RV (Continuous Case) :

$$\Delta_n^{-1/2} (RV - IV) \xrightarrow{d} N\left(0, 2 \int_0^1 c_s^2 ds\right)$$

Equivalently, writing $\sigma_s = \sqrt{c_s}$ (so $c_s = \sigma_s^2$ and $c_s^2 = \sigma_s^4$) :

$$\Delta_n^{-1/2} (RV - IV) \xrightarrow{d} N\left(0, 2 \int_0^1 \sigma_s^4 ds\right)$$

11 Jump Returns and the Truncated Variance

In practice, asset prices can exhibit sudden large moves-**jumps**-driven by earnings announcements, macroeconomic news, or market microstructure events. We model log-prices as :

$$dX_t = \sqrt{c_t} dW_t + dJ_t$$

where J_t is a pure jump process. The process decomposes as :

$$X_t = X_t^c + X_t^d$$

where X_t^c is the continuous (diffusive) part and X_t^d is the discontinuous (jump) part. the drift part is negligible at small time periods.

The Problem with RV Under Jumps As established above, under this model :

$$RV \xrightarrow{p} IV + \sum_p |\Delta_{\tau_p} X|^2$$

RV *overestimates* IV when jumps are present, since it absorbs the squared jump sizes. A jump-robust estimator is needed.

11.1 The Bipower Variance

The **bipower variance** (BV) is defined as :

$$BV = \frac{\pi}{2} \sum_{i=2}^n |r_i r_{i-1}|$$

where $r_i = \Delta_i^n X$ are intraday log-returns, and the constant $\frac{\pi}{2}$ is a bias correction factor arising from the expected absolute value of a standard normal : $E[|Z|] = \sqrt{2/\pi}$, so $(E[|Z|])^{-2} = \frac{\pi}{2}$.

Why BV is Jump-Robust The key insight is that jumps are rare. In practice, equity markets experience roughly 15-20 large jumps per year. When sampling at 5-minute intervals (78 per day), the probability of observing two jumps in adjacent 5-minute intervals is essentially zero.

More formally, the product $|r_i r_{i-1}|$ involves the returns from two *adjacent* intervals. If interval i contains a jump, the adjacent interval $i - 1$ almost surely does not (given jump sparsity). Therefore :

$$\frac{\pi}{2} |r_{t,i} r_{t,i-1}| = \frac{\pi}{2} |(r_{t,i}^c + r_{t,i}^d)(r_{t,i-1}^c + r_{t,i-1}^d)| \approx \frac{\pi}{2} |r_{t,i}^c \cdot r_{t,i-1}^c|$$

Since at most one of $\{r_{t,i}^d, r_{t,i-1}^d\}$ is non-zero at a time, the cross-terms involving r^d vanish. Thus :

$$BV \xrightarrow{p} IV$$

Efficiency Comparison

- **When there are no jumps :** RV is more efficient than BV (lower asymptotic variance).
- **When there are jumps :** RV is inconsistent (does not converge to IV), while BV remains consistent.
- BV is consistent but not the most efficient jump-robust estimator-its asymptotic variance is approximately $2.61 \times \int c_s^2 ds$, larger than the minimum $2 \int c_s^2 ds$.

11.2 Separating Diffusive Returns from Jump returns

Given intraday returns $r_{t,i}$, we want to classify each return as :

$$r_{t,i} = r_{t,i}^c + r_{t,i}^d \quad (\text{diffusive} + \text{jump})$$

If there are *no* jumps over the interval $[t_i - \Delta_n, t_i]$, the return is approximately :

$$r_{t,i} \approx \sigma_{t,i} Z_{t,i}, \quad Z_{t,i} \stackrel{iid}{\sim} N(0, 1), \quad \sigma_{t,i}^2 = \int_{t_i - \Delta_n}^{t_i} c_s ds$$

Using properties of the normal distribution :

- $P(|r_{t,i}| \leq 3\sigma_{t,i}) \approx 99.73\%$
- $P(|r_{t,i}| \leq 4\sigma_{t,i}) \approx 99.99\%$

Therefore : If $|r_{t,i}| > 4\sigma_{t,i}$ (or some multiple $\alpha\sigma_{t,i}$), it is almost certain that a jump occurred in interval i .

$$|r_{t,i}| \leq \alpha_n \sigma_{t,i} : \quad \text{return is } \mathbf{diffusive} \quad (r_{t,i} = r_{t,i}^c)$$

$$|r_{t,i}| > \alpha_n \sigma_{t,i} : \quad \text{return is a } \mathbf{jump} \quad (r_{t,i} = r_{t,i}^d)$$

where α_n is a threshold multiplier (in practice, $\alpha \in [3.5, 4.5]$).

The Problem : $\sigma_{t,i}$ is Unobservable The local volatility $\sigma_{t,i}$ is not directly observable. We need to estimate it. The BV estimator provides an estimate of the *daily* integrated variance IV_t . To get the local variance at interval i , we need to “redistribute” the daily variance to each interval.

The Diurnal Pattern (Intraday Seasonality)

Volatility is not uniform throughout the trading day. It is higher at the open (9 :30 am) and close (4 :00 pm), and lower around lunchtime. This **diurnal** or **intraday seasonal** pattern must be accounted for, otherwise the threshold will be too tight at the open/close and too loose at lunchtime.

We introduce a **time-of-day factor** τ_i to capture this pattern. The local variance approximation becomes :

$$r_{t,i} \overset{d}{\approx} N(0, \Delta_n \tau_i IV_t)$$

Computing τ_i

Step 1 : Compute the average bipower product at each intraday interval i across all T days :

$$b_i = \frac{1}{T} \sum_{t=1}^T |r_{t,i} r_{t,i-1}|, \quad i = 2, \dots, n, \quad b_1 \equiv b_2$$

Step 2 : Normalise to obtain the time-of-day factor :

$$\tau_i = \frac{b_i}{\frac{1}{n} \sum_{j=1}^n b_j}$$

This ensures that $\frac{1}{n} \sum_{i=1}^n \tau_i = 1$, so τ_i represents the relative (not absolute) intraday variance. Intervals at the open/close have $\tau_i > 1$; lunchtime intervals have $\tau_i < 1$.

The standard deviation of a diffusive return is $\Delta_n^{0.5} \sqrt{\tau_i IV_t}$. Naively, the threshold would be $\alpha \cdot \Delta_n^{0.5} \sqrt{\tau_i IV_t}$. However, as $\Delta_n \rightarrow 0$, this threshold also shrinks to zero, and *all* returns (including diffusive ones) would eventually be classified as jumps (since returns shrink as $\sqrt{\Delta_n}$ but the threshold shrinks at the same rate-the balance is delicate).

To ensure the threshold decreases *slower* than the diffusive returns as $\Delta_n \rightarrow 0$, we inflate it by :

$$\Delta_n^{-0.01} = \left(\frac{1}{n}\right)^{-0.01} = n^{0.01} > 1$$

This gives the **cutoff** :

$$\text{cutoff}_{t,i} = \alpha \cdot \Delta_n^{0.49} \cdot \sqrt{\tau_i BV_t}$$

where BV_t replaces the unknown IV_t , and the exponent 0.49 (rather than 0.50) embodies the inflation factor. Any exponent strictly less than 0.50 but close to it works in theory.

The Classification Rule in Practice With the cutoff defined, the diffusive and jump returns are :

$$r_{t,i}^c = r_{t,i} \cdot \mathbf{1}_{\{|r_{t,i}| \leq \alpha \Delta_n^{0.49} \sqrt{\tau_i BV_t}\}}$$

$$r_{t,i}^d = r_{t,i} \cdot \mathbf{1}_{\{|r_{t,i}| > \alpha \Delta_n^{0.49} \sqrt{\tau_i BV_t}\}}$$

where $\alpha \in [3.5, 4.5]$.

11.3 The Truncated Variance

Now that we can separate diffusive from jump returns, we construct the **truncated variance** (TV) :

The **truncated variance** for day t is :

$$TV_t = \sum_{i=1}^n \left(\Delta_{i+(t-1)n}^n X \right)^2 \mathbf{1}_{\{|\Delta_{i+(t-1)n}^n X| \leq \text{cutoff}_{t,i}\}} = \sum_{i=1}^n (r_{t,i}^c)^2$$

It sums the squared returns *only* for those intervals classified as diffusive.

Central Limit Theorem for Truncated Variance :

Under regularity conditions (which allow jumps in X_t) :

$$\Delta_n^{-1/2} (TV_t - IV_t) \overset{d}{\rightarrow} N\left(0, 2 \int_{t-1}^t c_s^2 ds\right)$$

This is the same asymptotic distribution as for RV in the *no-jump* case, but TV achieves this even when jumps are present. In case of BV,the limiting variance is 2.61 times the integral part. It has higher variance than TV. So, it is not efficient.

Jump Robustness : TV is consistent for IV even when the price process contains jumps. This is the primary advantage over RV. BV and TV are robust to jumps, while RV not.

Efficiency : The asymptotic variance $2 \int c_s^2 ds$ is the **minimum achievable variance** (the Cramér-Rao lower bound in this context) for estimating IV from high-frequency data. No other estimator can have a smaller asymptotic variance. TV is therefore the **most efficient** jump-robust estimator.

Comparison with BV : BV is also jump-robust and consistent, but its asymptotic variance is :

$$\text{Avar}(BV) \approx 2.61 \times \int_{t-1}^t c_s^2 ds$$

which is roughly 2.61 times larger than the minimum. TV achieves the minimum, making it considerably more efficient.

11.4 Comparison of IV Estimators

Estimator	Formula	Consistency	Robustness	Asym var factor
RV	$\sum r_i^2$	Yes (no jumps)	No	$2 \int c_s^2 ds$
BV	$\frac{\pi}{2} \sum r_i r_{i-1} $	Yes	Yes	$2.61 \times \int c_s^2 ds$
TV	$\sum r_i^2 \cdot \mathbf{1}_{\{ r_i \leq \text{cutoff}\}}$	Yes	Yes	$2 \int c_s^2 ds$

The Role of BV in the TV Procedure Although TV dominates BV in asymptotic efficiency, **BV is indispensable in computing TV** : BV is used to construct the cutoff, $\text{cutoff}_{t,i} = \alpha \Delta_n^{0.49} \sqrt{\tau_i BV_t}$. At the stage of classification, TV itself is unknown, so BV serves as the preliminary estimator. Once diffusive returns are identified using BV, they are fed into the TV formula to obtain the efficient estimate.

In practice,

- **Always use TV** as the primary estimate of daily IV. It is consistent under jumps and achieves the lowest asymptotic variance.
- **Use BV** as a robustness check or diagnostic : if $TV_t \ll RV_t$, this suggests significant jump activity on day t .
- **Use RV** only if you have strong a priori reasons to believe the process is continuous (no jumps), in which case it achieves the same efficiency as TV.
- The test statistic $RV_t/BV_t - 1$ (suitably normalised) can be used as a formal jump test.

Overview

1. We want to estimate $IV_t = \int_{t-1}^t c_s ds$, the daily integrated variance.
2. **Realized Variance (RV)** : $RV_t = \sum r_{t,i}^2$ is the natural first estimator. Under no jumps, $RV_t \xrightarrow{P} IV_t$ (LLN), and $\sqrt{n}(RV_t - IV_t) \xrightarrow{d} N(0, 2 \int c_s^2 ds)$ (CLT).
3. **The Jump Problem** : When jumps occur, $RV_t \xrightarrow{P} IV_t + \sum_p |\Delta_{\tau_p} X|^2 \neq IV_t$.
4. **Bipower Variance (BV)** : Replacing adjacent-pair products with absolute values kills the jump contamination. $BV_t \xrightarrow{P} IV_t$ under jumps. Needed as an input for jump detection.
5. **Jump Detection** : Using the cutoff $\alpha \Delta_n^{0.49} \sqrt{\tau_i BV_t}$, returns are classified as diffusive or jump. The diurnal factor τ_i accounts for intraday volatility seasonality.
6. **Truncated Variance (TV)** : Summing only squared diffusive returns gives $TV_t \xrightarrow{P} IV_t$ under jumps, with the *same* optimal CLT as RV under no-jump- $\sqrt{n}(TV_t - IV_t) \xrightarrow{d} N(0, 2 \int c_s^2 ds)$.

The quantity $\int_0^1 c_s^2 ds$ must itself be estimated to form confidence intervals. This can be done using the **realised quacticity** :

$$RQ = \frac{n}{3} \sum_{i=1}^n (\Delta_i^n X)^4 \xrightarrow{P} \int_0^1 c_s^2 ds$$

(under no jumps, a truncated version handles the jump case).

$$\text{Log-price process : } dX_t = \sqrt{c_t} dW_t + dJ_t$$

$$\text{Integrated Variance : } IV_t = \int_{t-1}^t c_s ds$$

$$\text{Realized Variance : } RV_t = \sum_{i=1}^n r_{t,i}^2 \xrightarrow{P} IV_t + \sum_p |\Delta_{\tau_p} X|^2$$

$$\text{Bipower Variance : } BV_t = \frac{n}{n-1} \frac{\pi}{2} \sum_{i=2}^n |r_{t,i}| |r_{t,i-1}| \xrightarrow{P} IV_t$$

$$\text{Time-of-day factor : } \tau_i = \frac{b_i}{\frac{1}{n} \sum_{j=1}^n b_j}, \quad b_i = \frac{1}{T} \sum_{t=1}^T |r_{t,i} r_{t,i-1}|$$

$$\text{Cutoff : } \text{cutoff}_{t,i} = \alpha \Delta_n^{0.49} \sqrt{\tau_i BV_t}, \quad \alpha \in [3.5, 4.5]$$

$$\text{Truncated Variance : } TV_t = \sum_{i=1}^n r_{t,i}^2 \mathbf{1}_{\{|r_{t,i}| \leq \text{cutoff}_{t,i}\}} \xrightarrow{P} IV_t$$

$$\text{CLT (RV, no jumps) : } \Delta_n^{-1/2} (RV - IV) \xrightarrow{d} N\left(0, 2 \int_0^1 c_s^2 ds\right)$$

$$\text{CLT (TV, with jumps) : } \Delta_n^{-1/2} (TV_t - IV_t) \xrightarrow{d} N\left(0, 2 \int_{t-1}^t c_s^2 ds\right)$$

12 Inference for the Integrated Variance

The foundation of Integrated variance is based on the **jump-diffusion model** for log-asset prices. The log-price process X_t is decomposed into a continuous (diffusive) component and a discontinuous (jump) component as,

$$X_t = X_t^c + X_t^d$$

The return of asset at day t and intraday interval i is written as :

$$r_{t,i} = r_{t,i}^c + r_{t,i}^d$$

where $r_{t,i}^c$ is the continuous (diffusive) return and $r_{t,i}^d$ is the jump return. Using the **bipower variance estimator** (BV), we can separate these two components.

The Truncated Variance Estimator : Using only the continuous returns, we estimate the Integrated Variance (IV) using the Truncated Variance (TV) estimator, which is jump-robust :

$$TV_t = \sum_{i=1}^n (r_{t,i}^c)^2$$

The asymptotic theory shows that for large n (equivalently, as $\Delta_n \rightarrow 0$) :

$$\Delta_n^{-0.50} (TV_t - IV_t) \overset{d}{\sim} N\left(0, 2 \int_{t-1}^t c_s^2 ds\right) \quad (1)$$

where c_s is the **spot/local variance** process (instantaneous variance at time s). To use this limiting distribution for inference, we need to estimate its asymptotic variance $2 \int_{t-1}^t c_s^2 ds$.

12.1 Quartic Integrated Variation (QIV)

The asymptotic variance from (1) involves the integral of the squared local variance. This is called the **Quartic Integrated Variation** :

$$QIV_t \equiv \int_{t-1}^t c_s^2 ds$$

The QIV is estimated by the following sum of fourth-power continuous returns :

$$\widehat{QIV}_t \equiv (3\Delta_n)^{-1} \sum_{i=1}^n (r_{t,i}^c)^4$$

To understand why this normalisation is necessary, recall that the continuous return can be approximated as :

$$r_{t,i}^c = \sqrt{\int_{t_i-\Delta_n}^{t_i} c_s ds} \cdot Z_{t,i}$$

where $Z_{t,i} \sim N(0,1)$ are i.i.d. standard normal random variables (arising from the independent increments of Brownian motion).

Assuming $c_s = c$ is constant for simplicity :

$$r_{t,i}^c = \sqrt{\Delta_n} c Z_{t,i}$$

Now, consider the sum of fourth-power returns :

$$\sum_{i=1}^n \left(r_{t,i}^c\right)^4 = \sum_{i=1}^n \Delta_n^2 c^2 Z_{t,i}^4 = c^2 \Delta_n \times \Delta_n \sum_{i=1}^n Z_{t,i}^4$$

By the **Law of Large Numbers** (LLN) : since $Z_{t,1}^4, Z_{t,2}^4, \dots, Z_{t,n}^4$ are i.i.d. (continuous transformations of i.i.d. normals are still i.i.d.), we have :

$$\Delta_n \sum_{i=1}^n Z_{t,i}^4 \rightarrow E[Z^4] = 3$$

using the well-known fact that the fourth moment of a standard normal is 3.

However, the factor $c^2 \Delta_n \rightarrow 0$, so without normalisation :

$$\sum_{i=1}^n \left(r_{t,i}^c\right)^4 \rightarrow 0$$

which is useless. By multiplying by $(3\Delta_n)^{-1}$:

$$(3\Delta_n)^{-1} \sum_{i=1}^n \left(r_{t,i}^c\right)^4 = \frac{c^2}{3} \left(\Delta_n \sum_{i=1}^n Z_{t,i}^4 \right) \rightarrow c^2$$

In the general case where c_s is time-varying, this extends to :

$$\widehat{QIV}_t \rightarrow \int_{t-1}^t c_s^2 ds = QIV_t$$

confirming consistency.

12.2 Confidence Intervals for the Integrated Variance

There are two primary approaches to construct confidence intervals for IV_t : (1) the asymptotic theory approach and (2) the bootstrap method.

Approach 1 : Using the Asymptotic Distribution

Since $\widehat{QIV}_t$ consistently estimates QIV_t , we can standardise the CLT in (1) :

$$\Delta_n^{-\frac{1}{2}} \frac{TV_t - IV_t}{\sqrt{2\widehat{QIV}_t}} \stackrel{d}{\sim} N(0,1)$$

This directly gives us a $100(1-\alpha)\%$ confidence interval for IV_t :

$$IV_t \in \left[TV_t + q_Z(\alpha/2) \sqrt{\Delta_n 2\widehat{QIV}_t}, \quad TV_t + q_Z(1-\alpha/2) \sqrt{\Delta_n 2\widehat{QIV}_t} \right]$$

where $q_Z(\alpha/2)$ is the $\alpha/2$ quantile of the standard normal (e.g. $q_Z(2.5\%) = -1.96$ for a 5% test).

We can further simplify by expanding the square root expression :

$$\sqrt{\Delta_n 2\widehat{QIV}_t} = \sqrt{2\Delta_n \cdot \frac{1}{3\Delta_n} \sum_{i=1}^n \left(r_{t,i}^c\right)^4} = \sqrt{\frac{2}{3} \sum_{i=1}^n \left(r_{t,i}^c\right)^4}$$

The final confidence interval simplifies to :

$$IV_t \in \left[TV_t + q_Z(\alpha/2) \sqrt{\frac{2}{3} \sum_{i=1}^n \left(r_{t,i}^c\right)^4}, \quad TV_t + q_Z(1-\alpha/2) \sqrt{\frac{2}{3} \sum_{i=1}^n \left(r_{t,i}^c\right)^4} \right]$$

Interpreting the Confidence Interval : A 95% confidence level means, if we were to repeat the sampling procedure 1000 times and compute an interval each time, approximately 950 of those 1000 intervals would contain the true unknown IV_t . For any *single* realised dataset, we cannot determine whether our specific interval contains the truth.

Approach 2 : The Bootstrap Method The bootstrap is a data-driven alternative to deriving the asymptotic distribution analytically. The idea :

1. Treat the observed data as the population.
2. Draw random samples *with replacement* to form “new” samples.
3. Compute TV on each new sample.
4. Use the empirical distribution of these estimates to form a confidence interval.

This is especially useful when the asymptotic variance is hard to derive analytically. However, it is computationally intensive.

The Blockwise Bootstrap : Respecting Heteroskedasticity A crucial subtlety in high-frequency data is the **diurnal (intraday) pattern** : volatility is highest at market open and close, and lowest in the middle of the day. If we draw returns randomly from anywhere in the day, we violate this structure.

Solution : Blockwise Bootstrap.

Fix $T = 1$ (one trading day). Partition the n continuous returns $r_1^c, r_2^c, \dots, r_n^c$ into M non-overlapping *blocks* of length k_n , such that $Mk_n = n$:

$$\underbrace{r_1^c, \dots, r_{k_n}^c}_{\text{1st block}}, \quad \underbrace{r_{k_n+1}^c, \dots, r_{2k_n}^c}_{\text{2nd block}}, \quad \dots, \quad \underbrace{r_{(M-1)k_n+1}^c, \dots, r_{Mk_n}^c}_{\text{Mth block}}$$

To generate one bootstrap sample, draw k_n returns *with replacement from each block separately* :

$$\underbrace{\tilde{r}_1^c, \dots, \tilde{r}_{k_n}^c}_{\text{1st resampled block}}, \quad \underbrace{\tilde{r}_{k_n+1}^c, \dots, \tilde{r}_{2k_n}^c}_{\text{2nd resampled block}}, \quad \dots$$

By drawing only within each block, we preserve the intraday volatility pattern.

12.2.1 Computing the Bootstrap Estimator and Confidence Interval

For each bootstrap sample, compute :

$$\widetilde{TV} = \sum_{i=1}^{Mk_n} (\tilde{r}_i^c)^2$$

Repeat $B = 10,000$ times to obtain $BS(TV) \equiv \{\widetilde{TV}_1, \dots, \widetilde{TV}_{10000}\}$.

The $100(1-\alpha)\%$ bootstrap confidence interval is :

$$CI(TV, \alpha) \equiv [q_{BS(TV)}(\alpha/2), q_{BS(TV)}(1-\alpha/2)]$$

where $q_{BS(TV)}(p)$ denotes the p -quantile of the bootstrap distribution. For a 95% CI, use the 2.5% and 97.5% quantiles.

12.2.2 Comparing the Two Approaches

	Asymptotic	Bootstrap
Speed	Fast	Slow
Theory required	Derive limiting distribution	Verify bootstrap validity
Computation	Low	High (10,000 resamples per day)
Quality	Same	Same

Bootstrap confidence intervals are neither better nor worse than asymptotic ones ; they are equivalent in quality when valid. In the high-frequency setting, the bootstrap does work, so both approaches are valid.

12.3 Confidence Intervals for the Annualised Integrated Variance

In practice, daily variance estimates are typically reported as annualised figures (e.g. annualised volatility). To derive confidence intervals for such transformations, we use the **Delta Method**.

The Delta Method Suppose $\hat{\theta}_n$ is an estimator of θ with limiting distribution :

$$\sqrt{n} (\hat{\theta}_n - \theta) \stackrel{d}{\rightarrow} N(0, \sigma^2)$$

Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a function differentiable at θ . Then :

$$\sqrt{n} (g(\hat{\theta}_n) - g(\theta)) \stackrel{d}{\rightarrow} N(0, [g'(\theta)]^2 \sigma^2)$$

$\hat{\theta}$ will be TV_t or BV_t and the θ will be IV_t . Starting from the no-jump CLT (using RV instead of TV for notational simplicity) :

$$\Delta_n^{-\frac{1}{2}} (RV_t - IV_t) \stackrel{d}{\sim} N\left(0, 2 \int_{t-1}^t c_s^2 ds\right)$$

Note : $\Delta_n^{-1/2} = \sqrt{n}$.

Define the annualisation transformation :

$$g(x) = \sqrt{252 \times x}$$

(Since there are 252 trading days per year and because our target is to find the confidence interval of variance using the standard deviation, we choose this form of function)

By the Delta Method :

$$\Delta_n^{-\frac{1}{2}} \left(\sqrt{252 RV_t} - \sqrt{252 IV_t} \right) \stackrel{d}{\sim} N(0, [g'(IV_t)]^2 \cdot 2 QIV_t)$$

Replacing population quantities with consistent estimators ($QIV_t \leftarrow \widehat{QIV}_t$ and $IV_t \leftarrow TV_t$) :

$$\Delta_n^{-\frac{1}{2}} \frac{\sqrt{252 RV_t} - \sqrt{252 IV_t}}{g'(TV_t) \sqrt{2\widehat{QIV}_t}} \stackrel{d}{\sim} N(0,1)$$

This statistic can be inverted to produce confidence intervals for the true annualised volatility $\sqrt{252 IV_t}$.

13 Estimating the Local Variance

We have estimated the **integrated variance** (variance over a whole day). We now turn to estimating the **local (spot) variance** c_t : the instantaneous variance at a specific time t within the trading day.

The log-price process follows :

$$dX_t = \sqrt{c_t} dW_t + dJ_t$$

where c_t is the local variance process we wish to estimate. With $T = 1$ (one trading day), let $i_t \in \mathbb{N}$ denote the interval index such that :

$$t \in ((i_t - 1)\Delta_n, i_t\Delta_n]$$

For example, with 5-minute intervals, if $t = 14:26:10$, then t lies in $(14:25:00, 14:30:00]$ and i_t identifies that interval. We cannot estimate c_t at the exact second, but we can estimate it for the interval containing t .

The analysis covers three cases :

1. c_t is continuous and t is in the interior of the day.
2. c_t is continuous but t is near the start or end of the day.
3. c_t is discontinuous at t (the local variance itself jumps).

13.1 Continuous Variance process and Time in the middle of the Day

13.1.1 The Local Variance Estimator

Lets assume c_t is a continuous process and fix some time t we are interested in. The idea for estimating ct is to construct a local version of the integrated variance estimator, where we sum up the squared continuous returns around the time t and divide by the size of the intervals

$$\hat{c}_t \equiv \frac{1}{(2k_n + 1)\Delta_n} \sum_{j=-k_n}^{k_n} \left(r_{i_t+j}^c\right)^2$$
$$\sum_{j=-k_n}^{k_n} \left(r_{i_t+j}^c\right)^2 \approx \int_{(i_t-1-k_n)\Delta_n}^{(i_t+k_n)\Delta_n} c_s ds$$

Change of variables. Set $s = t + s'$, centring around t . Since $i_t\Delta_n - t \in (0, \Delta_n]$ is negligible when Δ_n is small :

$$\sum_{j=-k_n}^{k_n} \left(r_{i_t+j}^c\right)^2 \approx \int_{(-k_n-1)\Delta_n}^{k_n\Delta_n} c(t+s) ds$$
$$\frac{1}{b+a} \int_{-a}^b f(x+s) ds = \frac{F(x+b) - F(x-a)}{(x+b) - (x-a)}$$

Taking limits $a, b \rightarrow 0$, the right-hand side converges to $F'(x) = f(x)$.

Conclusion : Applying this with $f = c$, $x = t$, $b = k_n \Delta_n$, $a \approx k_n \Delta_n$:

$$\hat{c}_t \approx \frac{1}{(2k_n + 1)\Delta_n} \int_{(-k_n-1)\Delta_n}^{k_n\Delta_n} c(t+s) ds \rightarrow c(t) \quad \text{as } k_n\Delta_n \rightarrow 0$$

This convergence is analogous to local regression smoothing : averaging a function over a shrinking neighbourhood consistently estimates the function’s value at that point.

13.2 Choosing the Parameter k_n

The parameter k_n controls the width of the local window. There is a fundamental **bias-variance tradeoff** :

- **Need $k_n \rightarrow \infty$** : To sum over many terms so that the variance of $\hat{c}_t$ goes to zero (precision). The variance is proportional to $1/k_n$.
- **Need $k_n\Delta_n \rightarrow 0$** : So the window shrinks to zero and $\hat{c}_t$ hones in on the true c_t (consistency/bias).

These two requirements must be balanced simultaneously. The **optimal choice** is :

$$k_n = K \Delta_n^{-1/2} = K\sqrt{n}$$

where K is a constant chosen by common sense.

$$k_n = K\sqrt{n} \rightarrow \infty$$
$$k_n\Delta_n = K\Delta_n^{1/2} = \frac{K}{\sqrt{n}} \rightarrow 0$$

With 5-minute data, $n \approx 77$, so $\sqrt{n} \approx 8.77$. The **rule of thumb** suggests keeping k_n between 5 and 13. Sensitivity checks should confirm that the estimator is stable across this range.

13.3 Continuous Variance and Time at the Beginning/End of the Day

When t is close to market open (9 :30) or close (16 :00), there may not be enough intervals on one side of i_t to use a full window of k_n .

Key constraint : Do *not* use returns from the previous or next trading day. The overnight period (16 :00 to 9 :30) often contains major news (earnings releases, macro announcements) that dramatically changes volatility dynamics. Using overnight returns would contaminate the local variance estimate.

Solution : Adjust the summation limits to stay within $\{1, 2, \dots, n\}$:

$$\hat{c}_t \equiv \frac{1}{N_t \Delta_n} \sum_{j=-J_1}^{J_2} (r_{t_i+j}^c)^2$$

where $J_1, J_2 \geq 0$ are chosen so that $1 \leq i_t + j \leq n$ for all j in the sum, and $N_t = J_1 + J_2 + 1$ is the total number of terms used.

For example, if $t \approx 15:47$ (close to end of day), we might only have $J_2 = 2$ forward intervals (until 16 :00) but $J_1 = k_n = 5$ backward intervals. The estimator uses only those available observations.

13.4 Discontinuous Variance Process

Just as prices can jump, so too can the local variance process c_t . If c_t has a discontinuity at time t , we estimate the **left** and **right** limits separately.

Define :

$$c_t^- \equiv \lim_{\tau \nearrow t} c_\tau \quad (\text{left limit}) \quad c_t^+ \equiv \lim_{\tau \searrow t} c_\tau \quad (\text{right limit})$$

Since c_t is discontinuous, $c_t^- \neq c_t^+$ (the variance can jump in either direction).

The left-limit estimator uses only returns *before* interval i_t :

$$\hat{c}_t^- \equiv \frac{1}{k_n\Delta_n} \sum_{j=1}^{k_n} (r_{t_i-j}^c)^2$$

The right-limit estimator uses only returns *after* interval i_t :

$$\hat{c}_t^+ \equiv \frac{1}{k_n\Delta_n} \sum_{j=1}^{k_n} (r_{t_i+j}^c)^2$$

The same optimal bandwidth $k_n = K\sqrt{n}$ applies, and the same boundary adjustments from Case 2 apply when t is near the start or end of the day. Once we have $\hat{c}_t^-$ and $\hat{c}_t^+$, we can take a weighted average to obtain a final estimate of c_t in the jump scenario, or report both limits to characterise the jump in variance.

Connection to the Bipower Variance : This three-case structure for the local variance mirrors the structure for integrated variance : just as we used the bipower variance to handle jumps in the price process, we now use one-sided estimators to handle jumps in the variance process.

13.5 Jump Inference

The local variance estimator $\hat{c}_t$ has an important further application : it enables statistical inference about individual jump sizes. Once we detect a jump in the return series (using truncation or bipower methods), a natural question is, *how large is this jump, and is it statistically significant ?* This matters for trading : a large, significant jump may be a strong signal, while a small one may be within normal diffusive variation.

13.5.1 The Jump as a Parameter and Its Estimator

For the interval $\tilde{j}$ that contains a jump at time τ , the full return decomposes as :

$$\Delta_{\tilde{j}}^n X = \Delta_{\tilde{j}}^n X^c + \Delta_\tau X$$

The **jump size** $J_\tau \equiv \Delta_\tau X$ is the true parameter of interest. Our estimator is simply the observed interval return :

$$\hat{J}_\tau = \Delta_{\tilde{j}}^n X$$

The estimation error is the diffusive return over the interval :

$$\hat{J}_\tau - J_\tau = \Delta_{\tilde{j}}^n X^c = r_j^c$$

Using our standard approximation for the continuous return :

$$\Delta_{\tilde{j}}^n X^c \approx N(0, \Delta_n c_\tau)$$

where c_τ is the local variance at the time of the jump. After standardising :

$$\frac{\Delta_n^{-1/2} (\hat{J}_\tau - J_\tau)}{\sqrt{c_\tau}} \approx N(0, 1)$$

Before the local variance material, the above was infeasible since c_τ was unknown. Now, replacing c_τ with $\hat{c}_\tau$:

$$\frac{\Delta_n^{-1/2} (\hat{J}_\tau - J_\tau)}{\sqrt{\hat{c}_\tau}} \approx N(0, 1)$$

A 95% confidence interval for the true jump size J_τ is :

$$\left[\hat{J}_\tau - 1.96 \Delta_n^{1/2} \sqrt{\hat{c}_\tau}, \quad \hat{J}_\tau + 1.96 \Delta_n^{1/2} \sqrt{\hat{c}_\tau} \right]$$

Since the local variance itself may jump at τ , one should be careful about which version of $\hat{c}_\tau$ to use. If c_s is continuous at τ (only the price jumps, not the variance), the standard estimator $\hat{c}_t$ (centred around τ) is appropriate. If c_s is also discontinuous at τ , one might use $\hat{c}_\tau^-$ or $\hat{c}_\tau^+$ depending on the precise timing.

Integrated Variance : $IV_t = \sum_{i=1}^n (r_{t,i}^c)^2$ (Jump-robust via truncation)

Quartic Variation : $\widehat{IQ}_t = \frac{1}{3\Delta_n} \sum_{i=1}^n (r_{t,i}^c)^4$ (Estimator for Integrated Quarticity)

Local Variance : $\hat{c}_t = \frac{1}{(2k_n + 1)\Delta_n} \sum_{j=-k_n}^{k_n} (r_{t+j}^c)^2, \quad k_n = K\sqrt{n}$

Left Limit : $\hat{c}_t^- = \frac{1}{k_n\Delta_n} \sum_{j=1}^{k_n} (r_{t-j}^c)^2$

Right Limit : $\hat{c}_t^+ = \frac{1}{k_n\Delta_n} \sum_{j=1}^{k_n} (r_{t+j}^c)^2$

Jump Size : $\hat{J}_\tau = \Delta X_\tau = X_\tau - X_{\tau-}$

Bootstrap CI : $[q_{BS}(\alpha/2), q_{BS}(1 - \alpha/2)]$

Jump CI : $\hat{J}_\tau \pm z_{1-\alpha/2} \sqrt{\Delta_n(\hat{c}_\tau^- + \hat{c}_\tau^+)}$

14 Consumption Based Asset Pricing

Agents maximize the trade off variance and expected returns in the CAPM model. Agents will care about consumption not about the mean and variance of returns. So, agents capture smoothing desire by investing today and if tomorrow the condition is bad, the investment will be used to smooth out his consumption.

Assumptions and Notations

- Agent has a concave utility function. More wealthy, more consumption then utility will increase. No market friction.
- Choose an optimal allocation between consumption and investment. The agent can consume now or invest in n securities.
- $c_t, \tilde{c}_{t+1}$: agent consumption, tilda indicates that it is random in the next time period.
- $y_t, \tilde{y}_{t+1}$: the agent’s income
- θ : n -vector with position in assets.
- p : n -vector of asset prices at time t .
- $\tilde{x}$: n -vector of asset pay-offs at time $t + 1$
- δ : subjective discount factor of future consumption (penalty for postponing consumption).

14.1 Agent’s Two Period Model

- Consumption now, $u(c_t)$
- Future consumption is uncertain, $\mathbb{E}[u(c_{t+1})]$
- Total utility, $U(c_t, c_{t+1}) = u(c_t) + \delta \mathbb{E}[u(c_{t+1})]$
- Investment and consumption at time t : $c_t = y_t - \theta' p$
- Consumption at time $t + 1$: $\tilde{c}_{t+1} = \tilde{y}_{t+1} + \theta' \tilde{x}$
- θ_i can be positive or negative. Negative is when shorting.

$$\max u(c_t) + \delta \mathbb{E}[u(\tilde{c}_{t+1})]$$

$$s.t \quad c_t = y_t - \theta' p \quad \tilde{c}_{t+1} = \tilde{y}_{t+1} + \theta' \tilde{x}$$

Substitute the constraints, and derive w.r.t θ and substitute back $c_t, \tilde{c}_{t+1}$ we get,

$$u'(c_t)p = \delta E[u'(\tilde{c}_{t+1})\tilde{x}]$$

This is called **Euler Equation**

- $u'(c_t)p_i$: loss of current utility by buying an extra asset i .
- $\mathbb{E}[\delta u'(\tilde{c}_{t+1})\tilde{x}_i]$: discounted gain in expected future utility.
- Equilibrium if left and right hand side are equal.

Marginal utility $u(c_t)'$ is high, when the agent is poor, meaning any consumption is too expensive for him. Low marginal utility is a good state for the agent.

14.2 Expected utility and Asset Pricing

Optimal investment and consumption leads to Euler equation. Current marginal costs equal future expected marginal gains.

$$pu'(c_t) = E[\delta u'(\tilde{c}_{t+1})\tilde{x}]$$

. The consequences for the asset pricing are that Euler equation defines the stochastic discount factor, also called pricing kernal.

14.3 Prices as a Function of pay-offs and Optimal consumption

$$u'(c_t)p = \delta E[u'(\tilde{c}_{t+1})\tilde{x}] \Leftrightarrow p = E[\frac{\delta u'(\tilde{c}_{t+1})}{u'(c_t)}\tilde{x}]$$

- Prices are expected discounted pay-offs

- The term $\delta u'(\tilde{c}_{t+1})/u'(c_t)$ is the discount factor, often denoted by $\tilde{m}$
- $\tilde{m}$ is a random variable, so it's called the stochastic discount factor (SDF) and pricing kernel
- Its realized value depends on the state of the world that occurs at $t + 1$. If it is highest in a situation it means agents value extra consumption most.

14.4 Stochastic Discount Factor

$$\tilde{m} = \frac{\delta u'(\tilde{c}_{t+1})}{u'(c_t)} \quad p = E[\tilde{m}\tilde{x}]$$

- $\tilde{m}$:marginal rate of substitution of current consumption for future consumption. $\tilde{m}$ discounts pay-offs into prices.
- **Good state** - Consumption : High, Utility : High, Marginal utility : low, m_s : low
- **Bad state** - Consumption : low, Utility : low, Marginal utility : high, m_s : high

14.5 Returns

$$p_i = E[\tilde{m}\tilde{x}] \Leftrightarrow E[\tilde{m}\frac{\tilde{x}_i}{p_i}] = E[\tilde{m}\tilde{R}_i] = 1$$

- Expected discounted gross returns equal 1. Gross return, $\tilde{R}_i$
- $\tilde{m}$ corrects for risk ; after risk correction, expected returns equal 1 $1 = E[\tilde{m}\tilde{R}_i] = \mathbb{E}[\tilde{m}\tilde{R}_j]$ does not imply $\mathbb{E}[\tilde{R}_i] = E[\tilde{R}_j]$

- $\tilde{m}$ and $\tilde{R}_i$ can be correlated, so $E[\tilde{m}\tilde{R}_i] \neq E[\tilde{m}]E[\tilde{R}_i]$

14.6 Excess Returns

$$\begin{aligned} E[\tilde{m}\tilde{R}^e] &= E[\tilde{m}(\tilde{R}_i - \tilde{R}_j)] = E[\tilde{m}\tilde{R}_i] - E[\tilde{m}\tilde{R}_j] \\ &= 1 - 1 = 0 \end{aligned}$$

- $\tilde{R}^e$: Excess return. Expected discounted excess returns equal 0, $E[\tilde{m}\tilde{R}^e] = 0$
- $\tilde{R}_j$, can be risk-free rate, but not necessarily

14.7 Risk Free Rate

Return R_f is certain, known at time t :

$$1 = E[\tilde{m}R_f] = E[\tilde{m}]R_f \Leftrightarrow E[\tilde{m}] = \frac{1}{R_f}$$

$$R_f = \frac{1}{E[\tilde{m}]} = \frac{1}{\delta} \frac{u'(c_t)}{E[u'(\tilde{c}_{t+1})]}$$

- Gross risk free rate, R_f and kernel pricing are inverse to each other
- Future consumption increases, future marginal utility decreases, risk free rate increases

14.8 Relation to Factor Models

Result in Asset pricing is the equivalence between a factor model and a stochastic discount factor, is there exist $a \in R, b \in R^k$ such that $\tilde{m} = a + b'f$ is SDE, with $\mathbb{E}[\tilde{m}R_i] = 1 \forall i$ if and only if there exist $\lambda_0 \in R, \lambda \in R^k$ for every asset i

$$E[R_i] = \lambda_0 + \lambda' \beta_i \quad \beta_i = cov[f, f']^{-1} cov[f, R_i]$$

$$\lambda_0 = \frac{1}{E[\tilde{m}]} \text{ and } \lambda = -\frac{cov[f, f']b}{E[\tilde{m}]}$$

In consumption based asset pricing model, factor models look for variables that are good proxies for aggregate marginal utility growth, i.e. variables for which

$$m_{t+1} = \beta \frac{u'(\tilde{c}_{t+1})}{u'(c_t)} \approx a + b'f_{t+1}$$

15 General Moment Method

The General Moment Method is useful to find the different moments of the random variables. These moments help us to find the distribution of the population parameters, like first moment gives mean and second moment gives variance. Without limited data, we can estimate the moments and find the distribution of the population parameters.

z is a vector of observed random variables, g is q-vector of known fuctions that depend on data z and a true parameter $\theta_0 \in \Theta \subseteq R^p$. θ_0 satisfies the following,

$$\mathbb{E}[g(z, \theta)] = 0_q \quad : \text{Unconditional Moment of Equality mode}$$

- If there exist unique solution θ to UMEM, then it is called point identified. q = equations, p = unknowns
- If $q = p$, then θ is just identified. If $q > p$, then θ is over identified

15.1 GMM Estimation

Since we have only finite sample data, but not population data, we dont know $\mathbb{E}[g(z, \theta)] = 0$ so we estimate θ by considering moment condition as, $\frac{1}{n} \sum g(z, \theta) = 0$

- If $q = p$, then estimator $\hat{\theta}$ which solves these equations is called method of moments of θ
- If $q < p$, there are more unknowns than equations so we can't get any solution for θ , then we try to make $\frac{1}{n} \sum g(z, \theta)$ as close as zero using a particular metric.

$$\hat{\theta}_{Gmm} = \arg \min_{\theta \in \Theta} \left[\frac{1}{n} \sum_{i=1}^n g(z_i, \theta) \right]' w \left[\frac{1}{n} \sum_{i=1}^n g(z_i, \theta) \right]$$

w is a $q \times q$ positive definite matrix.To write concisely, we first define the sample moment conditions : $\bar{g}_n(\theta) = \frac{1}{n} \sum_{i=1}^n g(z_i, \theta)$ The objective function and the GMM estimator is the value that minimizes this function is defined as :

$$Q_n(\theta) = \bar{g}_n(\theta)' W \bar{g}_n(\theta) \quad \hat{\theta}_{\text{GMM}} = \arg \min_{\theta} Q_n(\theta)$$

15.2 Consistency and Asymptotic Normality

- By CLT, $\sqrt{n} \bar{g}_n(\theta_0) \xrightarrow{d} \mathcal{N}(0, \Omega)$
- By LLN, $\frac{\partial \bar{g}_n(\hat{\theta})}{\partial \theta'} \xrightarrow{p} G'$ and $\frac{\partial \bar{g}_n(\tilde{\theta})}{\partial \theta} \xrightarrow{p} G$
- Hence, $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} [G'WG]^{-1}G'W \cdot \mathcal{N}(0, \Omega)$

$$V = (G'WG)^{-1}G'WNG(G'WG)^{-1}$$

$$\Omega = E[g(z, \theta_0)g(z, \theta_0)'] \quad G = \mathbb{E} \left[\frac{\partial}{\partial \theta} g(z, \theta) \right]$$

We choose,

$$W_{opt} = E[g(z, \theta_0)g(z, \theta_0)']^{-1} = \Omega^{-1}$$

so that we get asymptotic variance of GMM estimator as $V_{opt} = (G'\Omega^{-1}G)^{-1}$. In practice we don't know W_{opt} but we can estimate it.

- Choose $W = I_q$ and obtain the initial GMM estimator $\tilde{\theta}_n$ using this weighting matrix. Since $\tilde{\theta}_n \xrightarrow{p} \theta_0$, we can then estimate the optimal weighting matrix $\hat{W}_{\text{opt}}$ by :

$$\hat{W}_{\text{opt}} = \left[\frac{1}{n} \sum_{i=1}^n g(z_i, \tilde{\theta}_n)g(z_i, \tilde{\theta}_n)' \right]^{-1}$$

- Re-estimate θ by solving the following minimization problem,

$$\hat{\theta}_n = \arg \min_{\theta \in \Theta} \bar{g}_n(\theta)' \hat{W}_{\text{opt}} \bar{g}_n(\theta)$$

The resulting estimator $\hat{\theta}_n$ is asymptotically optimal.

15.3 Conditional Moment Equality Models

$$\begin{aligned} \mathbb{E} \left[g(z, \theta_0) \middle| X = x \right] &= 0 \quad f(x) \mathbb{E} \left[g(z, \theta_0) \middle| X = x \right] = \mathbb{E} \left[f(x)g(z, \theta_0) \middle| X = x \right] = 0 \\ &\quad \mathbb{E} [f(x)g(z, \theta_0)] = 0 \end{aligned}$$

Example : From Linear Regresion

$$\mathbb{E}[u|X] = 0 \quad \Rightarrow \mathbb{E}[Xu] = 0 \quad \Rightarrow \mathbb{E}[X(Y - X\beta)] = 0$$

From Asset Pricing,

$$E[\mu_{t+1}R_{it+1}] = 1 \Rightarrow E[\mu_{t+1}R_{it+1} - 1] = 0$$

16 State Space Models and Filtering

A **state space model** is a flexible framework that separates a system into two layers, an unobservable **state** that evolves over time, and an **observable measurement** that is a noisy projection of that state. Almost any linear time series model AR, MA, ARMA, random walks can be written in this form, making it a unifying language for econometric modelling.

The fundamental motivation is **estimation**. In a standard AR(1) model we observe every lag directly, so writing down the likelihood is trivial. In a state space model the state $\mathbf{x}_t$ is *latent*. It drives what we observe ($\mathbf{z}_t$) but is not observed itself. To estimate the model's parameters we must first recover the conditional mean and variance of $\mathbf{z}_t$, and that is exactly what the **Kalman filter** provides.

16.1 State Space Representation

Every state space model consists of exactly two equations :

Transition equation (state dynamics) :

$$\mathbf{x}_{t+1} = \mathbf{F} \mathbf{x}_t + \mathbf{G} \mathbf{w}_{t+1}, \quad \mathbf{w}_{t+1} \sim \mathcal{N}(\mathbf{0}, \mathbf{Q})$$

Measurement equation (linking state to observables) :

$$\mathbf{z}_t = \mathbf{H}' \mathbf{x}_t + \mathbf{v}_t, \quad \mathbf{v}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$$

The noise processes $\{\mathbf{w}_t\}$ and $\{\mathbf{v}_t\}$ are mutually independent and serially independent.

The key variable is $\mathbf{x}_t$, the **state vector**, it is unobserved. The variable $\mathbf{z}_t$ is the **observed measurement vector**. The two noise processes are assumed to be mutually independent and independent of the past.

- $\mathbf{x}_t$ ($n \times 1$) : Latent state vector (unobserved)
- $\mathbf{z}_t$ ($m \times 1$) : Observable measurement vector
- $\mathbf{F}$ ($n \times n$) : Transition (companion) matrix
- $\mathbf{G}$ ($n \times k$) : Shock loading matrix
- $\mathbf{H}$ ($n \times m$) : Measurement loading matrix
- $\mathbf{w}_t$ ($k \times 1$) : State noise ; $Cov(\mathbf{w}_t) = \mathbf{Q}$
- $\mathbf{v}_t$ ($m \times 1$) : Measurement noise ; $Cov(\mathbf{v}_t) = \mathbf{R}$

The transition equation looks like a vector AR(1) process, the crucial difference is that $\mathbf{x}_t$ is **unknown**. The measurement equation then tells us how the observable $\mathbf{z}_t$ relates to that unknown state, subject to noise $\mathbf{v}_t$.

Once we can estimate the conditional distribution of $\mathbf{z}_t$ -its mean and covariance given past observations-we can write down and maximise the **likelihood function** to estimate all unknown parameters ($\mathbf{F}, \mathbf{G}, \mathbf{H}, \mathbf{Q}, \mathbf{R}$).

16.2 Non-Uniqueness of the State Space Representation

A subtle but important theoretical point : the state space representation of any given time series model is **not unique**. There exist infinitely many equivalent representations.

Basis Transformations Let $\boldsymbol{B}$ be any non-singular square matrix conformable with $\boldsymbol{F}$. Define a new state vector $\boldsymbol{x}_t^* = \boldsymbol{B}\boldsymbol{x}_t$ and transformed matrices :

$$\begin{aligned}\boldsymbol{F}^* &= \boldsymbol{B}\boldsymbol{F}\boldsymbol{B}^{-1}, \\ \boldsymbol{G}^* &= \boldsymbol{B}\boldsymbol{G}, \\ \boldsymbol{H}^{*'} &= \boldsymbol{H}'\boldsymbol{B}^{-1}.\end{aligned}$$

Then the system

$$\begin{aligned}\boldsymbol{x}_{t+1}^* &= \boldsymbol{F}^*\boldsymbol{x}_t^* + \boldsymbol{G}^*\boldsymbol{w}_{t+1}, \\ \boldsymbol{z}_t &= \boldsymbol{H}^{*'}\boldsymbol{x}_t^* + \boldsymbol{v}_t,\end{aligned}$$

is an entirely equivalent state space representation because

$$\boldsymbol{H}^{*'}\boldsymbol{x}_t^* = \boldsymbol{H}'\boldsymbol{B}^{-1} \cdot \boldsymbol{B}\boldsymbol{x}_t = \boldsymbol{H}'\boldsymbol{x}_t.$$

The observables $\boldsymbol{z}_t$ are unchanged.

Writing a model in state space form is a *trial-and-error process*. There is no unique correct answer. The goal is simply to find *any* combination of a transition equation and a measurement equation that together reproduce the original time series model when simplified. As soon as one valid representation is found, infinitely many more exist via basis transformations.

AR(2) Process - Example Consider the second-order autoregression :

$$z_t = \rho_1 z_{t-1} + \rho_2 z_{t-2} + v_t, \qquad v_t \sim \mathcal{N}(0, \sigma_v^2).$$

State vector : $\boldsymbol{x}_t = [y_t, \rho_2 y_{t-1}]'$

Transition equation :

$$\boldsymbol{x}_t = \begin{bmatrix} \rho_1 & 1 \\ \rho_2 & 0 \end{bmatrix} \boldsymbol{x}_{t-1} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} v_t$$

Measurement equation :

$$z_t = \begin{bmatrix} 1 & 0 \end{bmatrix} \boldsymbol{x}_t$$

State vector : $\boldsymbol{x}_t = [y_t, y_{t-1}]'$

Transition equation :

$$\boldsymbol{x}_t = \begin{bmatrix} \rho_1 & \rho_2 \\ 1 & 0 \end{bmatrix} \boldsymbol{x}_{t-1} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} v_t$$

Measurement equation :

$$z_t = \begin{bmatrix} 1 & 0 \end{bmatrix} \boldsymbol{x}_t$$

These two representations are related by the basis transformation $\boldsymbol{B} = \begin{bmatrix} 1 & 0 \\ 0 & \rho_2 \end{bmatrix}$ applied to Representation II, which recovers Representation I.

Verification (Representation I) : Expanding the first transition row gives

$$y_t = \rho_1 y_{t-1} + 1 \cdot (\rho_2 y_{t-2}) + v_t.$$

Combined with the measurement equation $z_t = y_t$, this recovers the original AR(2) exactly.

MA(1) Process - Example Consider the first-order moving average :

$$z_t = v_t + \theta v_{t-1}, \qquad v_t \sim \text{iid } \mathcal{N}(0, \sigma_v^2).$$

Key insight : the state vector can include *error terms*-a crucial demonstration of the framework's flexibility.

State vector : $\boldsymbol{x}_t = [y_t, \theta v_t]'$

Transition equation :

$$\boldsymbol{x}_t = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \boldsymbol{x}_{t-1} + \begin{bmatrix} 1 \\ \theta \end{bmatrix} v_t$$

Measurement equation :

$$z_t = \begin{bmatrix} 1 & 0 \end{bmatrix} \boldsymbol{x}_t$$

State vector : $x_t = v_{t-1}$ (scalar)

Transition equation : $x_t = v_{t-1}$

Measurement equation : $z_t = \theta x_t + v_t$

Notice the trade-off : Representation II has a trivial transition equation but a more complex measurement equation. This illustrates the general principle that *complexity can be shifted between the two equations*.

Random Walk with Drift (Non-Stationary Process) - Example The state space representation is not limited to stationary processes. Consider :

$$z_t = z_{t-1} + \beta + v_t, \qquad v_t \sim \mathcal{N}(0, \sigma_v^2).$$

This process has a **unit root** and a constant drift parameter β . It is

non-stationary :

$$\begin{aligned}\mathbb{E}[z_t \mid z_0] &= \beta t + z_0 \quad (\text{grows with } t), \\ \text{Var}(z_t \mid z_0) &= \sigma_v^2 t \quad (\text{grows with } t).\end{aligned}$$

State vector : $\boldsymbol{x}_t = [y_t, \beta]'$

Transition equation :

$$\boldsymbol{x}_t = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \boldsymbol{x}_{t-1} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} v_t$$

Measurement equation :

$$z_t = \begin{bmatrix} 1 & 0 \end{bmatrix} \boldsymbol{x}_t$$

The drift parameter β is included as an element of the state vector. The Kalman filter will then *estimate* it as part of its recursive procedure. Even fixed parameters can be embedded in the state-a further demonstration of the framework's extraordinary flexibility.

16.3 Stability Conditions

In this module we restrict attention to **stable systems**. Stability ensures that the Kalman filter converges to a unique steady state and that the initial conditions can be set in a principled, model-consistent way.

A state space system is **stable** if, for any initial state $\boldsymbol{x}_0$, the state vector $\boldsymbol{x}_t$ converges to a unique fixed point $\boldsymbol{x}^*$ as $t \rightarrow \infty$ (in the absence of shocks).

Necessary and Sufficient Condition :

$$|\lambda_i(\boldsymbol{F})| < 1 \qquad \text{for all eigenvalues } \lambda_i \text{ of } \boldsymbol{F}.$$

This condition is directly analogous to the stationarity condition for an AR(p) process, which requires that all roots of the characteristic polynomial lie outside the unit circle.

The random walk example violates this condition : the $\boldsymbol{F}$ matrix $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ has eigenvalues $\lambda = 1$ (with multiplicity 2), so $|\lambda| = 1 \not< 1$. Non-stationary models require special treatment (diffuse priors, etc.) beyond the scope of this module.

17 Kalman Filter

The **Kalman filter** is a recursive algorithm that solves the state estimation problem optimally. It processes observations $z_1, z_2, \dots, z_T$ sequentially, updating our best estimate of the latent state at each step.

Symbol	Definition	Interpretation
$\boldsymbol{x}_{t t-1}$	$\mathbb{E}(\boldsymbol{x}_t \mid \boldsymbol{z}^{t-1})$	Best guess of $\boldsymbol{x}_t$ using info up to $t-1$
$\boldsymbol{x}_{t t}$	$\mathbb{E}(\boldsymbol{x}_t \mid \boldsymbol{z}^t)$	Best guess of $\boldsymbol{x}_t$ using info up to t (updated)
$\boldsymbol{z}_{t t-1}$	$\boldsymbol{H}'\boldsymbol{x}_{t t-1}$	Predicted measurement at time t
$\boldsymbol{x}_{t+1 t}$	$\boldsymbol{F}\boldsymbol{x}_{t t}$	
$\boldsymbol{\Sigma}_{t t-1}$	$\text{Var}(\boldsymbol{x}_t \mid \boldsymbol{z}^{t-1})$	State prediction error variance
$\boldsymbol{\Sigma}_{t t}$	$\text{Var}(\boldsymbol{x}_t \mid \boldsymbol{z}^t)$	State filtered error variance
$\boldsymbol{\Omega}_{t t-1}$	$\text{Var}(\boldsymbol{z}_t \mid \boldsymbol{z}^{t-1})$	Measurement prediction error variance

Suppose we are at time t . We already have :

- Our best state estimate $\boldsymbol{x}_{t|t-1}$ from the previous step
- The associated uncertainty $\boldsymbol{\Sigma}_{t|t-1}$
- A new observation $\boldsymbol{z}_t$ just arrived

The Kalman filter asks : how should we **optimally update** $\boldsymbol{x}_{t|t-1}$ to incorporate the new information in $\boldsymbol{z}_t$, yielding $\boldsymbol{x}_{t|t}$? The answer has a beautifully intuitive form, a **weighted average** of the prior estimate and the new information :

$$\boldsymbol{x}_{t|t} = \boldsymbol{x}_{t|t-1} + \boldsymbol{K}_t(\boldsymbol{z}_t - \boldsymbol{z}_{t|t-1}) = \boldsymbol{x}_{t|t-1} + \boldsymbol{K}_t(\boldsymbol{z}_t - \boldsymbol{H}'\boldsymbol{x}_{t|t-1})$$

where $\boldsymbol{K}_t$ is the **Kalman gain**, it measures how much we update $\boldsymbol{x}_{t|t-1}$ as a function in our error in predicting $\boldsymbol{z}_t$:

$$\boldsymbol{K}_t = \boldsymbol{\Sigma}_{t|t-1} \boldsymbol{H} \left(\boldsymbol{H}'\boldsymbol{\Sigma}_{t|t-1} \boldsymbol{H} + \boldsymbol{R} \right)^{-1}$$

The term $(\boldsymbol{z}_t - \boldsymbol{H}'\boldsymbol{x}_{t|t-1})$ is the **innovation** or **prediction error**-the new information arriving at time t that was not anticipated. The Kalman gain $\boldsymbol{K}_t$ determines how strongly we revise our estimate in response to this surprise.

Intuition of the Kalman Gain : The formula for $\boldsymbol{K}_t$ makes deep intuitive sense :

- $\boldsymbol{\Sigma}_{t|t-1}$ (**state uncertainty**) : If we are very uncertain about $\boldsymbol{x}_t$, we should trust the new observation more $\Rightarrow$ large $\boldsymbol{K}_t$.
- $\boldsymbol{R}$ (**measurement noise**) : If the observation $\boldsymbol{z}_t$ is very noisy, we should trust it less $\Rightarrow$ large $\boldsymbol{R}$ reduces $\boldsymbol{K}_t$.
- $\boldsymbol{H}$ links the two : it determines how well $\boldsymbol{z}_t$ reflects the state.

Numerical example. Consider the simple model $x_t = \mu + v_t$ and $z_t = x_t + \xi_t$ with $\sigma_\xi^2 = q \sigma_v^2$ and $\Sigma_{1|0} = \sigma_v^2$:

$$K_1 = \frac{1}{1+q}.$$

When q is large (noisy measurement), $K_1 \rightarrow 0$: we barely update upon seeing z_1 . When $q \rightarrow 0$ (precise measurement), $K_1 \rightarrow 1$: we set our estimate equal to z_1 .

17.1 Probabilistic Approach

Suppose the joint distribution of $(\boldsymbol{X}, \boldsymbol{Y})$ conditional on information $\boldsymbol{W}$ is :

$$\begin{bmatrix} \boldsymbol{X} \\ \boldsymbol{Y} \end{bmatrix} \Big| \boldsymbol{W} \sim \mathcal{N} \left(\begin{bmatrix} \boldsymbol{x}^* \\ \boldsymbol{y}^* \end{bmatrix}, \begin{bmatrix} \boldsymbol{\Sigma}_{xx} & \boldsymbol{\Sigma}_{xy} \\ \boldsymbol{\Sigma}_{yx} & \boldsymbol{\Sigma}_{yy} \end{bmatrix} \right).$$

Then the conditional distribution of $\boldsymbol{X}$ given $\boldsymbol{Y} = \boldsymbol{y}$ and $\boldsymbol{W}$ is :

$$\boldsymbol{X} \mid \boldsymbol{y}, \boldsymbol{W} \sim \mathcal{N} \left(\boldsymbol{x}^* + \boldsymbol{\Sigma}_{xy} \boldsymbol{\Sigma}_{yy}^{-1} (\boldsymbol{y} - \boldsymbol{y}^*), \quad \boldsymbol{\Sigma}_{xx} - \boldsymbol{\Sigma}_{xy} \boldsymbol{\Sigma}_{yy}^{-1} \boldsymbol{\Sigma}_{yx} \right). \tag{2}$$

To apply this to the Kalman filter, identify :

- $\boldsymbol{X} \leftrightarrow \boldsymbol{x}_t$ (the latent state we want to estimate)
- $\boldsymbol{Y} \leftrightarrow \boldsymbol{z}_t$ (the new observation)
- $\boldsymbol{W} \leftrightarrow \boldsymbol{z}^{t-1}$ (all past observations)
- $\boldsymbol{x}^* = \boldsymbol{x}_{t|t-1}$, $\boldsymbol{\Sigma}_{xx} = \boldsymbol{\Sigma}_{t|t-1}$

To use the Lemma, we need $\boldsymbol{\Sigma}_{xx}$, $\boldsymbol{\Sigma}_{yy}$, and $\boldsymbol{\Sigma}_{xy}$ evaluated conditionally on $\boldsymbol{z}^{t-1}$.

Conditional mean of $\boldsymbol{z}_t$

$$\boldsymbol{z}_{t|t-1} = \mathbb{E}(\boldsymbol{z}_t \mid \boldsymbol{z}^{t-1}) = \mathbb{E}(\boldsymbol{H}'\boldsymbol{x}_t + \boldsymbol{v}_t \mid \boldsymbol{z}^{t-1}) = \boldsymbol{H}'\boldsymbol{x}_{t|t-1}.$$

The measurement error $\boldsymbol{v}_t$ is serially independent and thus uncorrelated with $\boldsymbol{z}^{t-1}$, so $\mathbb{E}(\boldsymbol{v}_t \mid \boldsymbol{z}^{t-1}) = \boldsymbol{0}$.

Conditional variance of $\boldsymbol{z}_t$ ($\boldsymbol{\Sigma}_{yy}$)

$$\boldsymbol{\Omega}_{t|t-1} = \text{Var}(\boldsymbol{z}_t \mid \boldsymbol{z}^{t-1}) = \boldsymbol{H}'\boldsymbol{\Sigma}_{t|t-1} \boldsymbol{H} + \boldsymbol{R}.$$

This follows by writing $\boldsymbol{z}_t - \boldsymbol{z}_{t|t-1} = \boldsymbol{H}'(\boldsymbol{x}_t - \boldsymbol{x}_{t|t-1}) + \boldsymbol{v}_t$, squaring, taking conditional expectations, and using independence of $\boldsymbol{v}_t$ from $(\boldsymbol{x}_t - \boldsymbol{x}_{t|t-1})$:

- $\boldsymbol{H}'(\boldsymbol{x}_t - \boldsymbol{x}_{t|t-1})(\boldsymbol{x}_t - \boldsymbol{x}_{t|t-1})'\boldsymbol{H} \rightarrow \boldsymbol{H}'\boldsymbol{\Sigma}_{t|t-1} \boldsymbol{H}$
- $\boldsymbol{v}_t \boldsymbol{v}_t' \rightarrow \boldsymbol{R}$
- Cross-terms vanish because $\boldsymbol{v}_t \perp (\boldsymbol{x}_t - \boldsymbol{x}_{t|t-1})$

Cross-covariance ($\boldsymbol{\Sigma}_{xy}$)

$$\text{Cov}(\boldsymbol{z}_t, \boldsymbol{x}_t \mid \boldsymbol{z}^{t-1}) = \boldsymbol{H}'\boldsymbol{\Sigma}_{t|t-1}.$$

Expanding $\boldsymbol{z}_t - \boldsymbol{z}_{t|t-1}$, multiplying by $(\boldsymbol{x}_t - \boldsymbol{x}_{t|t-1})'$, and taking conditional expectations eliminates the $\boldsymbol{v}_t$ term (independent of $\boldsymbol{x}_t$ and $\boldsymbol{x}_{t|t-1}$), leaving $\boldsymbol{H}'\boldsymbol{\Sigma}_{t|t-1}$.

Applying the Lemma by substituting into the Conditional Normal Lemma (2) :

Updated mean ($\boldsymbol{x}_{t|t}$) :

$$\boldsymbol{x}_{t|t} = \boldsymbol{x}_{t|t-1} + \boldsymbol{\Sigma}_{t|t-1} \boldsymbol{H} \left(\boldsymbol{H}'\boldsymbol{\Sigma}_{t|t-1} \boldsymbol{H} + \boldsymbol{R} \right)^{-1} (\boldsymbol{z}_t - \boldsymbol{H}'\boldsymbol{x}_{t|t-1}) = \boldsymbol{x}_{t|t-1} + \boldsymbol{K}_t(\boldsymbol{z}_t - \boldsymbol{H}'\boldsymbol{x}_{t|t-1}).$$

Updated variance ($\Sigma_{t|t}$) :

$$\Sigma_{t|t} = \Sigma_{t|t-1} - \Sigma_{t|t-1} H \left(H' \Sigma_{t|t-1} H + R \right)^{-1} H' \Sigma_{t|t-1}.$$

The updated variance **always decreases** : incorporating new information $\mathbf{z}_t$ always reduces uncertainty about $\mathbf{x}_t$. The reduction is larger when the measurement is more precise (smaller $\mathbf{R}$) or when the state-observation covariance is larger.

17.2 Full Iteration Algorithm

The Kalman filter is a complete recursive algorithm. Each iteration consists of two phases : a prediction step (propagating the estimate forward) and an update step (incorporating the new observation).

Inputs at iteration t : $\mathbf{x}_{t|t-1}$, $\Sigma_{t|t-1}$, observation $\mathbf{z}_t$

- Prediction error variance

$$\Omega_{t|t-1} = H' \Sigma_{t|t-1} H + R$$

- Predicted measurement

$$\mathbf{z}_{t|t-1} = H' \mathbf{x}_{t|t-1}$$

- Update state estimate

$$\mathbf{x}_{t|t} = \mathbf{x}_{t|t-1} + \Sigma_{t|t-1} H \Omega_{t|t-1}^{-1} (\mathbf{z}_t - H' \mathbf{x}_{t|t-1})$$

- Update state variance

$$\Sigma_{t|t} = \Sigma_{t|t-1} - \Sigma_{t|t-1} H \Omega_{t|t-1}^{-1} H' \Sigma_{t|t-1}$$

- Predict next state

$$\mathbf{x}_{t+1|t} = \mathbf{F} \mathbf{x}_{t|t}$$

- Predict next state variance

$$\Sigma_{t+1|t} = \mathbf{F} \Sigma_{t|t} \mathbf{F}' + \mathbf{G} \mathbf{Q} \mathbf{G}'$$

- **Output** : $\mathbf{x}_{t+1|t}$, $\Sigma_{t+1|t}$ (inputs for iteration $t + 1$)

The filter’s power is its recursive structure. Starting from initial conditions ($\mathbf{x}_{1|0}$, $\Sigma_{1|0}$), a single pass through the data processes T observations in $O(T)$ time. At each step, the innovation $\boldsymbol{\nu}_t = \mathbf{z}_t - \mathbf{z}_{t|t-1}$ and its variance $\Omega_{t|t-1}$ are by-products used directly to construct the log-likelihood function.

17.3 Likelihood Function

The ultimate goal of running the Kalman filter is to enable **maximum likelihood estimation** (MLE) of the model parameters. The filter provides the conditional means and variances needed to evaluate the likelihood at every observation.

17.3.1 The Log-Likelihood

By the chain rule of probability, the joint likelihood of $\mathbf{z}^T = \{\mathbf{z}_t\}_{t=1}^T$ factorises as :

$$\log l(\mathbf{z}^T \mid \mathbf{F}, \mathbf{G}, \mathbf{H}, \mathbf{Q}, \mathbf{R}) = \sum_{t=1}^T \log l(\mathbf{z}_t \mid \mathbf{z}^{t-1}, \mathbf{F}, \mathbf{G}, \mathbf{H}, \mathbf{Q}, \mathbf{R}).$$

Since $\mathbf{z}_t \mid \mathbf{z}^{t-1} \sim \mathcal{N}(\mathbf{z}_{t|t-1}, \Omega_{t|t-1})$, each term is a multivariate normal log-density :

$$\log l = - \sum_{t=1}^T \left[\frac{N}{2} \log(2\pi) + \frac{1}{2} \log |\Omega_{t|t-1}| + \frac{1}{2} \boldsymbol{\nu}_t' \Omega_{t|t-1}^{-1} \boldsymbol{\nu}_t \right]$$

where :

$$\begin{aligned} \boldsymbol{\nu}_t &= \mathbf{z}_t - \mathbf{z}_{t|t-1} = \mathbf{z}_t - \mathbf{H}' \mathbf{x}_{t|t-1} && \text{(innovation)} \\ \Omega_{t|t-1} &= \mathbf{H}' \Sigma_{t|t-1} \mathbf{H} + \mathbf{R} && \text{(innovation variance)} \end{aligned}$$

Both quantities $\boldsymbol{\nu}_t$ and $\Omega_{t|t-1}$ are computed as by-products of the Kalman filter recursion. The log-likelihood is then maximised over the parameter space ($\mathbf{F}, \mathbf{G}, \mathbf{H}, \mathbf{Q}, \mathbf{R}$) using numerical optimisation to obtain the MLE.

For a standard AR(1) model $y_t = \rho y_{t-1} + \varepsilon_t$, the conditional mean ρy_{t-1} is directly observable. The Kalman filter plays the same role for state space models : it provides the conditional mean and variance of $\mathbf{z}_t$ when $\mathbf{x}_t$ is unobservable, making MLE feasible.

17.4 Initial Conditions

As a recursive algorithm, the Kalman filter requires **starting values** $x_{1|0}$ and $\Sigma_{1|0}$ to initialise the recursion. For stable systems, these are chosen as the **unconditional (steady-state) moments** of the system.

17.4.1 Initial State Mean

For a stable system, the state converges to a fixed point x^* satisfying :

$$x^* = F x^*.$$

For $|\lambda_i(F)| < 1$, this implies $x^* = \mathbf{0}$ (assuming the state has been demeaned). We set :

$$\mathbf{x}_{1|0} = \mathbf{0}.$$

17.4.2 Initial State Variance

The unconditional variance Σ^* of the state satisfies

$$\Sigma^* = \mathbf{F} \Sigma^* \mathbf{F}' + \mathbf{G} \mathbf{Q} \mathbf{G}'.$$

We set $\Sigma_{1|0} = \Sigma^*$ as the initial variance.

Steady-State Initialisation : By starting the filter at the system’s long-run equilibrium, we avoid imposing an arbitrary starting point that could distort the likelihood. The filter’s estimates will be consistent and asymptotically efficient. For short time series, the choice of initial conditions can matter, and robustness checks are advisable.

17.5 Diebold-Li Yield Curve Model

It models the entire yield curve, across all maturities simultaneously using just three latent factors. Estimation and forecasting are carried out via the Kalman filter.

Bond Pricing Setup : Let $P_t(\tau)$ denote the price at time t of a zero-coupon bond maturing in τ periods, and let $y_t(\tau)$ denote its continuously compounded yield :

$$P_t(\tau) = e^{-\tau y_t(\tau)}.$$

The yield $y_t(\tau)$ is the key variable. At any time t we observe yields across many maturities $\tau_1, \tau_2, \dots, \tau_N$. These yields **co-move** strongly across maturities, suggesting the existence of a small number of common latent factors.

17.5.1 The Diebold-Li Factor Model

For observation date t and maturity τ , the model specifies :

$$y_t(\tau) = L_t + S_t \left(\frac{1 - e^{-\lambda \tau}}{\lambda \tau} \right) + C_t \left(\frac{1 - e^{-\lambda \tau}}{\lambda \tau} - e^{-\lambda \tau} \right)$$

The three latent factors have clear economic interpretations :

- **Level (L_t)** : The factor loading is identically 1 for all maturities. $\lim_{\tau \rightarrow \infty} y_t(\tau) = L_t$. A 1-unit increase in L_t shifts the entire yield curve upward by 1 unit.
- **Slope (S_t)** : The loading $(1 - e^{-\lambda \tau})/(\lambda \tau)$ decays from 1 (at $\tau \rightarrow 0$) toward 0 (as $\tau \rightarrow \infty$). Empirically, S_t is highly negatively correlated with the spread $y_t(120\text{m}) - y_t(3\text{m})$ (10-year/3-month spread).
- **Curvature (C_t)** : The loading is hump-shaped-zero at both extremes and positive for medium maturities. C_t is correlated with $2 y_t(24\text{m}) - y_t(3\text{m}) - y_t(120\text{m})$, the standard definition of yield curve curvature.

The parameter λ controls where the slope loading achieves its maximum and is typically fixed (e.g. at $\lambda = 0.0609$ corresponding to a 30-month peak).

17.5.2 State Space Formulation

The Diebold-Li model becomes a state space system by specifying the factor dynamics as a **VAR(1)** :

Transition equation (factor dynamics) :

$$\begin{bmatrix} L_t - \mu_L \\ S_t - \mu_S \\ C_t - \mu_C \end{bmatrix} = \mathbf{A} \begin{bmatrix} L_{t-1} - \mu_L \\ S_{t-1} - \mu_S \\ C_{t-1} - \mu_C \end{bmatrix} + \begin{bmatrix} \eta_{Lt} \\ \eta_{St} \\ \eta_{Ct} \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Measurement equation (yield curve) :

$$\begin{bmatrix} y_t(\tau_1) \\ \vdots \\ y_t(\tau_N) \end{bmatrix} = \mathbf{\Lambda} \begin{bmatrix} L_t \\ S_t \\ C_t \end{bmatrix} + \begin{bmatrix} w_{\tau_1,t} \\ \vdots \\ w_{\tau_N,t} \end{bmatrix}$$

where the i -th row of the factor loading matrix $\mathbf{\Lambda}$ is :

$$\begin{bmatrix} 1 & \frac{1 - e^{-\lambda \tau_i}}{\lambda \tau_i} & \frac{1 - e^{-\lambda \tau_i}}{\lambda \tau_i} - e^{-\lambda \tau_i} \end{bmatrix}.$$

Here $\boldsymbol{\mu} = (\mu_L, \mu_S, \mu_C)'$ are the unconditional factor means. In practice, one redefines $\tilde{\mathbf{z}}_t = \mathbf{z}_t - \mathbf{\Lambda} \boldsymbol{\mu}$ to obtain a zero-mean system consistent with our general framework.

17.5.3 Yield Curve Shapes

Shape	Parameters	Description
Upward sloping (normal)	$L > 0, S < 0$	Long rates above short rates
Downward sloping (inverted)	$L > 0, S > 0$	Short rates above long rates and recession signal
Humped	$L > 0, S \approx 0, C > 0$	Medium-term rates highest
Flat	$L > 0, S \approx 0, C \approx 0$	All maturities yield similarly

Common yield curve shapes in the Diebold-Li framework

18 OLS Regression : Derivation & Properties

18.1 OLS Objective Function and First-Order Condition

Given a sample $y_1, \dots, y_n$, the OLS estimator minimises the **sum of squared residuals** :

$$Q(\hat{\beta}_0) = \sum_{i=1}^n (y_i - \hat{\beta}_0)^2$$

Setting the derivative to zero :

$$\frac{d}{d\hat{\beta}_0} \sum_{i=1}^n (y_i - \hat{\beta}_0)^2 = -2 \sum_{i=1}^n (y_i - \hat{\beta}_0) = 0$$

This yields the **intercept-only OLS estimate** :

$$\hat{\beta}_0 = \frac{1}{n} \sum_{i=1}^n y_i = \bar{y}$$

18.2 Simple Linear Regression (SLR) OLS Estimators

For the model $y_i = \beta_0 + \beta_1 x_i + u_i$:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

18.3 Variance of OLS Estimators

Population variance of the error :

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Estimated error variance (unbiased, using $n - 2$ degrees of freedom) :

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^n \hat{u}_i^2$$

where $\hat{u}_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$

Estimated variance and standard error of $\hat{\beta}_1$:

$$\widehat{\text{Var}}(\hat{\beta}_1) = \frac{\hat{\sigma}^2}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad \widehat{se}(\hat{\beta}_1) = \sqrt{\widehat{\text{Var}}(\hat{\beta}_1)}$$

18.4 Including Irrelevant Variables

If the true model is $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$ but we estimate with an *irrelevant* x_3 : OLS remains **unbiased** for β_1, β_2 , but **efficiency decreases** (variance of $\hat{\beta}$ increases).

18.5 Omitted Variable Bias

If the *true* model is $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$ but we estimate $y = \beta_0 + \beta_1 x_1 + v$, the bias in $\hat{\beta}_1$ has direction :

	$\text{Corr}(x_1, x_2) > 0$	$\text{Corr}(x_1, x_2) < 0$
$\beta_2 > 0$	Positive Bias	Negative Bias
$\beta_2 < 0$	Negative Bias	Positive Bias

18.6 Log-Linear Models

For the model $\ln(\text{price}) = \beta_0 + \beta_1 \ln(\text{nox}) + \beta_2 \text{rooms} + u$:

- β_1 is the **elasticity** of price with respect to nox (a 1% increase in nox $\Rightarrow \beta_1\%$ change in price).
- β_2 is the **semi-elasticity** (one extra room $\Rightarrow 100\beta_2\%$ change in price).

18.7 Variance of Error in Binary Outcome (LPM)

For $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ with binary $y_i \in \{0, 1\}$:

$$\mathbb{E}[\varepsilon_i \mid x_i] = 0,$$

$$\text{Var}(\varepsilon_i \mid x_i) = (\beta_0 + \beta_1 x_i)(1 - \beta_0 - \beta_1 x_i)$$

19 APT, CAPM, Sharpe Ratios & GRS Test

19.1 Arbitrage Pricing Theory (APT)

For a k -factor APT model the required return on a well-diversified portfolio p is :

$$E[R_p] = R_f + \beta_{p,1} \cdot RP_1 + \beta_{p,2} \cdot RP_2 + \dots + \beta_{p,k} \cdot RP_k$$

where RP_j is the **risk premium for factor j** .

Arbitrage portfolio construction. For a portfolio that is :

- Zero net investment** : $\sum_i w_i = 0$
- Zero factor exposure** : $\sum_i w_i \beta_{i,j} = 0$ for each factor j
- Positive expected profit** : if an asset is mispriced

19.2 CAPM

Under the CAPM, for any portfolio p :

$$E[R_p - R_f] = \beta_p E[R_M - R_f]$$

where $\beta_p = \frac{\text{Cov}(R_p, R_M)}{\text{Var}(R_M)}$

19.3 Sharpe Ratio and the Sharpe Ratio Relationship

The **Sharpe ratio** of a portfolio p :

$$SR_p = \frac{E[R_p - R_f]}{\sigma(R_p)}$$

Key result : When CAPM holds,

$$\frac{SR_p}{SR_M} = \frac{E[R_p - R_f]}{\sigma(R_p)} \cdot \frac{\sigma(R_M)}{E[R_M - R_f]} = \text{Corr}(R_M, R_p)$$

The **Sharpe ratio of the efficient tangency portfolio** :

$$Sh[\mathbf{r}] = \sqrt{\boldsymbol{\mu}' \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}}$$

19.4 GRS Test Statistic (Single Asset)

For the market-model regression $r_{it} = \alpha_i + \beta_i r_t^m + \epsilon_{it}$, the GRS t -statistic for a single asset is :

$$t_i = (T - 2) \left(1 + \frac{\hat{\mu}_m^2}{\hat{\sigma}_m^2} \right)^{-1} \frac{\hat{\alpha}_i^2}{\hat{\sigma}_i^2}$$

GRS as improvement in squared Sharpe ratio :

$$\frac{\hat{\alpha}_i^2}{\hat{\sigma}_i^2} = Sh[\mathbf{r}]^2 - Sh[r_m]^2$$

where $Sh[\mathbf{r}]^2 = \boldsymbol{\mu}' \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}$, computed from :

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_m \\ \alpha_i + \beta_i \mu_m \end{bmatrix}, \quad \boldsymbol{\Sigma} = \begin{bmatrix} \sigma_m^2 & \beta_i \sigma_m^2 \\ \beta_i \sigma_m^2 & \beta_i^2 \sigma_m^2 + \sigma_i^2 \end{bmatrix}, \quad \boldsymbol{\Sigma}^{-1} = \frac{1}{\sigma_m^2 \sigma_i^2} \begin{bmatrix} \beta_i^2 \sigma_m^2 + \sigma_i^2 & -\beta_i \sigma_m^2 \\ -\beta_i \sigma_m^2 & \sigma_m^2 \end{bmatrix}$$

19.5 AR(1) Model : Moments

For $\text{Return}_t = \beta_0 + \beta_1 \text{Return}_{t-1} + u_t$, $u_t \sim N(0, \sigma^2)$, $|\beta_1| < 1$:

$$E[\text{Return}_t] = \frac{\beta_0}{1 - \beta_1}$$

$$\text{Var}(\text{Return}_t) = \frac{\sigma^2}{1 - \beta_1^2}$$

19.6 F-Test for Joint Significance

To test $H_0 : \beta_1 = \beta_2 = 0$, use the **F-statistic** from the ANOVA table. Reject H_0 at significance level α if $p\text{-value} < \alpha$.

20 Cross-Sectional Regressions & Asset Pricing

20.1 Cross-Sectional OLS : Equal-Weighted Portfolio

For the regression $R_{it}^e = \delta_t + \epsilon_{it}$ ($i = 1, \dots, N$) :

$$\hat{\delta}_t = \frac{1}{N} \sum_{i=1}^N R_{it}^e = \bar{R}_t^e$$

20.2 Cross-Sectional OLS with Beta : Fama-MacBeth Slope

For $R_{it}^e = \theta_t + \lambda_t \beta_{it} + a_{it}$:

$$\hat{\lambda}_t = \frac{\sum_i (\beta_{it} - \bar{\beta}_t)(R_{it}^e - \bar{R}_t^e)}{\sum_i (\beta_{it} - \bar{\beta}_t)^2} = \frac{\sum_i (\beta_{it} - \bar{\beta}_t) R_{it}^e}{\sum_i (\beta_{it} - \bar{\beta}_t) \beta_{it}}$$

with weights :

$$w_{it}^\lambda = \frac{\beta_{it} - \bar{\beta}_t}{\sum_i (\beta_{it} - \bar{\beta}_t) \beta_{it}}, \quad \sum_i w_{it}^\lambda = 0, \quad \beta_t^\lambda = \sum_i w_{it}^\lambda \beta_{it} = 1$$

20.3 Probability Calculation Using Standard Normal

For i.i.d. normal R_i with mean μ and variance σ^2 :

$$P\left(\sum_{i=1}^4 R_i < 0\right) = \Phi\left(\frac{-4\mu}{\sqrt{4}\sigma}\right)$$

20.4 Multi-Factor Asset Pricing Model

A valid k -factor model satisfies :

$$E[R] - R_f = \boldsymbol{\lambda}'_F \text{Var}[\mathbf{F}]^{-1} \text{Cov}[\mathbf{F}, R]$$

Aggregating to a single factor $G = \mathbf{b}'\mathbf{F}$:

$$\mathbf{b} = \text{Var}[\mathbf{F}]^{-1} \boldsymbol{\lambda}_F$$

20.5 Momentum Effect Decomposition (Jegadeesh-Titman)

$$\mathbb{E}[(r_{it} - \bar{r}_t)(r_{it-1} - \bar{r}_{t-1})] = \sigma_\mu^2 + \sigma_\beta^2 \text{Cov}[f_t, f_{t-1}] + \widetilde{\text{Cov}}[e_{it}, e_{it-1}]$$

21 Poisson Processes & Factor Models

21.1 Poisson Process

A **Poisson process** with rate λ :

$$P(N(t) - N(s) = k) = e^{-\lambda(t-s)} \frac{[\lambda(t-s)]^k}{k!}, \quad k = 0, 1, 2, \dots$$

21.2 Interarrival and Arrival Times

Interarrival times T_n are i.i.d. $\text{Exp}(\lambda)$. **Arrival time of the n -th event** :

$$S_n = \sum_{i=1}^n T_i \sim \text{Gamma}(n, \lambda)$$

21.3 Uniform Distribution of Arrivals

Given $N(t) = 1$, the arrival time $S_1 \sim \text{Unif}(0, t)$:

$$P(S_1 \leq s \mid N(t) = 1) = \frac{s}{t}$$

21.4 Compound Poisson Process

Let $X_t = \sum_{i=1}^{N(t)} Y_i$ where Y_i are i.i.d. with mean μ_Y and variance σ_Y^2 .

$$E[X(t)] = \lambda t \mu_Y$$

$$\text{Var}[X(t)] = \lambda t (\sigma_Y^2 + \mu_Y^2) = \lambda t E[Y_i^2]$$

CLT for compound Poisson :

$$\frac{X(t) - \lambda t \mu_Y}{\sqrt{\lambda t (\sigma_Y^2 + \mu_Y^2)}} \xrightarrow{d} N(0, 1)$$

21.5 Fama-French Six-Factor Model

$$r_t^e = c + \beta_1 \text{MKTRF}_t + \beta_2 \text{SMB}_t + \beta_3 \text{HML}_t + \beta_4 \text{RMW}_t + \beta_5 \text{CMA}_t + \beta_6 \text{MOM}_t + \epsilon_t$$

22 Realised Volatility, Jumps & Microstructure Noise

22.1 Realised Variance

$$RV = \sum_{i=1}^n |\Delta_i^n X|^2 \stackrel{p}{\rightarrow} \int_0^1 c_s \, ds = IV \quad \text{as } n \rightarrow \infty$$

22.2 Jump Classification Threshold Rule

$$r_{ti}^c = r_{ti} \mathbf{1}_{\{|r_{ti}| \leq cut_{ti}\}}, \quad r_{ti}^d = r_{ti} \mathbf{1}_{\{|r_{ti}| > cut_{ti}\}}$$

$$cut_{ti} = \alpha \sqrt{\tau_i B V_t \Delta_n^{0.49}}$$

22.3 Microstructure Noise

Effect on RV :

$$\sum_{i=1}^n (Y_i^n - Y_{i-1}^n)^2 \approx IV + n(2\sigma_\chi^2)$$

First-order autocorrelation of noisy returns :

$$\text{Corr}(\Delta_i^n Y, \Delta_{i+1}^n Y) = \frac{-\sigma_\chi^2}{\Delta_n c + 2\sigma_\chi^2}$$

23 Asymptotic Theory, Bootstrap & Jump Confidence Intervals

23.1 Asymptotic Distribution of Realised Variance

$$\Delta_n^{-1/2} \left(\sum_{i=1}^n |\Delta_i^n X|^2 - \int_0^1 c_s \, ds \right) \stackrel{d}{\rightarrow} N(0, 2\Sigma)$$

where $\hat{\Sigma} = (3\Delta_n)^{-1} \sum_{i=1}^n (r_i^c)^4$.

23.2 Delta Method

If $\sqrt{n}(\hat{\theta}_n - \theta) \stackrel{d}{\rightarrow} N(0, \sigma^2)$ and g is differentiable at θ :

$$\sqrt{n}(g(\hat{\theta}_n) - g(\theta)) \stackrel{d}{\rightarrow} N(0, g'(\theta)^2 \sigma^2)$$

23.3 Bootstrap Confidence Interval for Truncated Variance

- Partition the n continuous returns into $M = \lfloor n/k_n \rfloor$ non-overlapping blocks of length k_n .
- In each bootstrap draw, resample k_n returns with replacement within each block ; concatenate to form a pseudo-sample.
- Compute bootstrap TV : $\widehat{TV} = \sum_{i=1}^{Mk_n} (\widehat{r}_i^c)^2$.
- Repeat 10,000 times ; form 95% CI using the 2.5th and 97.5th percentiles.

23.4 Confidence Interval for a Jump

95% confidence interval for jump size J_τ :

$$\left[\hat{J}_\tau - 1.96 \Delta_n^{1/2} \sqrt{\hat{c}_\tau}, \quad \hat{J}_\tau + 1.96 \Delta_n^{1/2} \sqrt{\hat{c}_\tau} \right]$$

24 Stochastic Discount Factor & Asset Pricing Theory

24.1 Stochastic Discount Factor (SDF) from Utility

$$m_{t+1} = \delta \frac{u'(c_{t+1})}{u'(c_t)}$$

Risk-free rate : $R_f = 1/E[m_{t+1}]$.

24.2 Asset Pricing Equation

$$\boxed{1 = E[m_{t+1} R_{t+1}]}$$
$$p_t = E[m_{t+1} x_{t+1}]$$

Risk premium : $E[R_i] - R_f = -R_f \text{Cov}(m_{t+1}, R_i)$.

24.3 Expected Utility Optimization : FOCs

$$U'(W - p^A \xi_A - p^B \xi_B) p^A = E[U'(\xi_A x_A + \xi_B x_B) x_A]$$
$$U'(W - p^A \xi_A - p^B \xi_B) p^B = E[U'(\xi_A x_A + \xi_B x_B) x_B]$$

24.4 Factor Pricing and Beta Representation

$$\boxed{E[R_i] = \lambda_0 + \boldsymbol{\lambda}' \boldsymbol{\beta}_i}$$
$$\text{where } \boldsymbol{\beta}_i = \text{Cov}[\mathbf{f}, \mathbf{f}']^{-1} \text{Cov}[\mathbf{f}, R_i]$$

24.5 CAPM as a Linear SDF

For CAPM, $M = a + bR_W$:

$$a = \frac{1}{R_f} - b E[R_W], \quad b = \frac{R_f - E[R_W]}{R_f \text{Var}[R_W]}$$

24.6 Hansen-Jagannathan Bound

$$\frac{\sigma(M)}{E[M]} \geq \frac{E[R - R_f]}{\sigma(R - R_f)}$$

25 Euler Equations, State Space Models & GMM

25.1 Euler Equation

$$u'(c_t) = E\big[\beta(1 + r_{t+1}) u'(c_{t+1})\big]$$

Moment condition under log utility :

$$E\left[\beta(1 + r_{t+1}) \frac{c_t}{c_{t+1}} - 1\right] = 0$$

25.2 Linearised SDF

$$\tilde{m}_t = \xi [1 - (z_t - \mu_z)], \quad \xi = \beta \exp(-\mu_z)$$

25.3 State Space Representation

$$x_{t+1} = F x_t + G w_{t+1} \quad z_t = A + H' x_t + v_t$$

MA(2) :

$$F = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad Q = \begin{bmatrix} \sigma^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad R = 0, \quad A = \mu, \quad H' = [1 \quad \theta_1 \quad \theta_2]$$

AR(p) :

$$F = \begin{bmatrix} \rho_1 & \cdots & \rho_{p-1} & \rho_p \\ 1 & \cdots & 0 & 0 \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & 1 & 0 \end{bmatrix}, \quad G = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad H' = [1 \quad 0 \quad \cdots \quad 0]$$

ARMA(1,1) :

$$F = \begin{bmatrix} \phi_1 & 1 \\ 0 & 0 \end{bmatrix}, \quad G = \begin{bmatrix} 1 \\ \theta \end{bmatrix}, \quad H' = [1 \quad 0]$$

25.4 GMM : Generalised Method of Moments

$$Q_{\mathbf{W}}(\theta) = g_T(\theta)' \mathbf{W} g_T(\theta)$$

Closed-form solution :

$$\hat{\mathbf{b}}_{\mathbf{W}} = \left\{ E_T[x_t \mathbf{R}'_t] \mathbf{W} E_T[\mathbf{R}_t x'_t] \right\}^{-1} E_T[x_t \mathbf{R}'_t] \mathbf{W} \mathbf{1}$$

Asymptotic distribution :

$$\sqrt{n}(\hat{\theta} - \theta_0) \stackrel{d}{\rightarrow} N(0, V) \quad V = (G'WG)^{-1}G'W\Omega WG(G'WG)^{-1}$$

Efficient GMM : Optimal weighting matrix $\mathbf{W}^* = \Omega^{-1}$, giving $V = (G'\Omega^{-1}G)^{-1}$.

25.5 Moment Conditions for MA(1) and AR(1)

MA(1) : $Y_t = \mu_0 + \epsilon_t + \theta_0 \epsilon_{t-1}$:

$$E[Y_t] = \mu_0, \quad \text{Var}(Y_t) = \sigma_0^2(1 + \theta_0^2), \quad \text{Cov}(Y_t, Y_{t-1}) = \sigma_0^2 \theta_0$$

AR(1) : $Y_t = c_0 + \phi_0 Y_{t-1} + \epsilon_t, |\phi_0| < 1$:

$$E[Y_t] = \frac{c_0}{1 - \phi_0}, \quad \text{Var}(Y_t) = \frac{\sigma_0^2}{1 - \phi_0^2}, \quad \text{Cov}(Y_t, Y_{t-1}) = \frac{\sigma_0^2 \phi_0}{1 - \phi_0^2}$$